UNIT 7
CLUSTER ANALYSIS

Elizabeth Wanner

About this material

- Materials by Yulan He / Felipe Campelo / Elizabeth Wanner, based on

  Tan, Steinbach and Kumar: ***Introduction to Data Mining: Lecture Notes*** (https://www-users.cs.umn.edu/~kumar001/dmbook/index.php)

  Al-Fuqaha's slide deck on Clustering Analysis (https://cs.wmich.edu/alfuqaha/summer14/cs6530/lectures/ClusteringAnalysis.pdf)

  M. Hahsler et al (2019), DBSCAN: Fast Density-Based Clustering with R. Journal of Statistical Software 91(1), 30pp.

- Required Reading:

  Tan et al. (2019): *Introduction to Data Mining*, 2nd ed., Chapter 5.

Topics

- What is Cluster Analysis
- Types of Clustering
  - Partitional: K-means
  - Density-based: DBSCAN
  - Hierarchical: Agglomerative Nesting
- Cluster Validity
  - Similarity Metric

Topics

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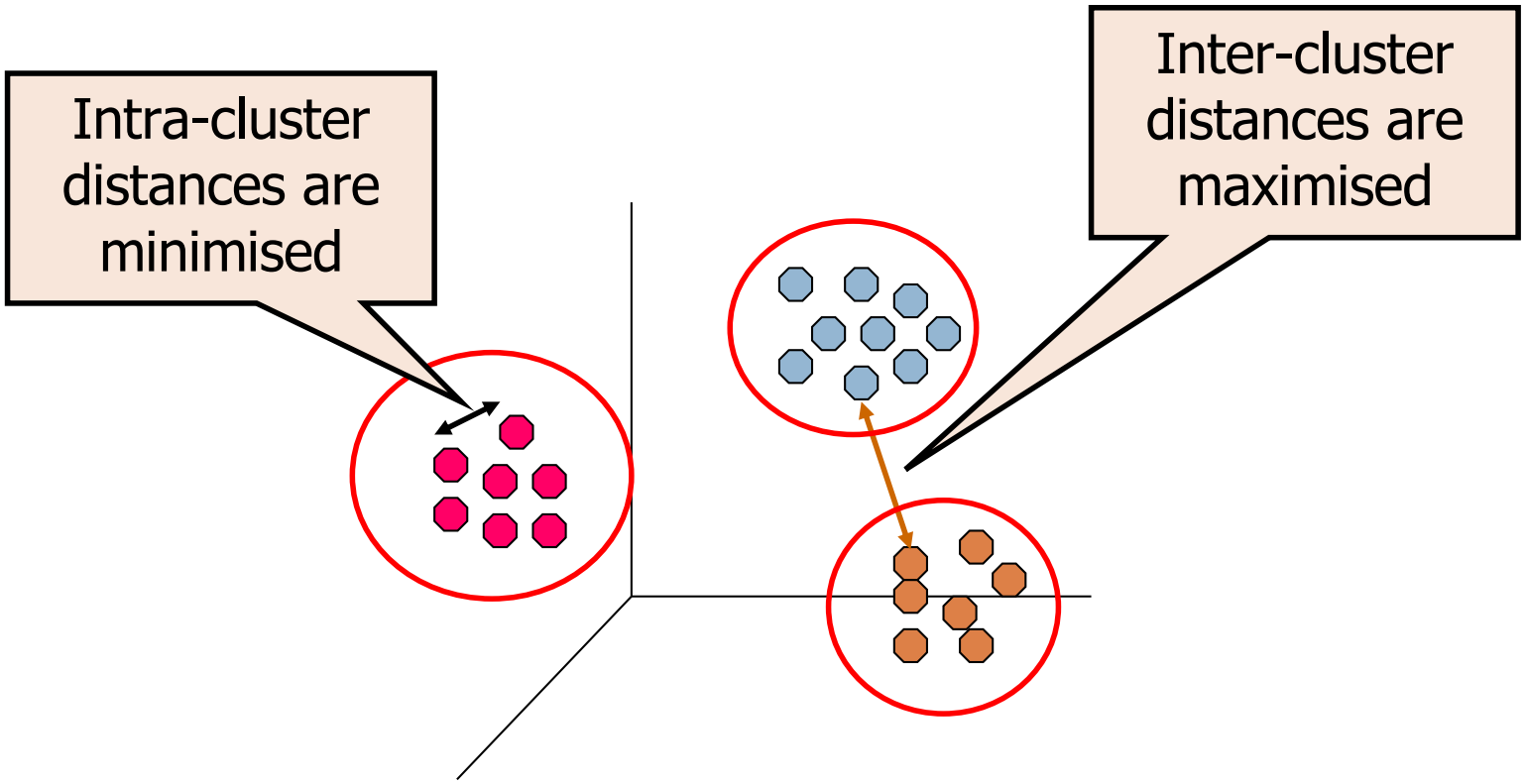
What is Cluster Analysis?

> *Cluster analysis groups data objects based only on information found in the data that describes the objects and their relationships. The goal is that the objects within a group be similar (or related) to one another and different from (or unrelated to) the objects in other groups. The greater the similarity (or homogeneity) within a group and the greater the difference between groups, the better or more distinct the clustering.*
>
> **Tan, Steinbach, Kumar, "Introduction to Data Mining", 2006**

What is Cluster Analysis?

- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups

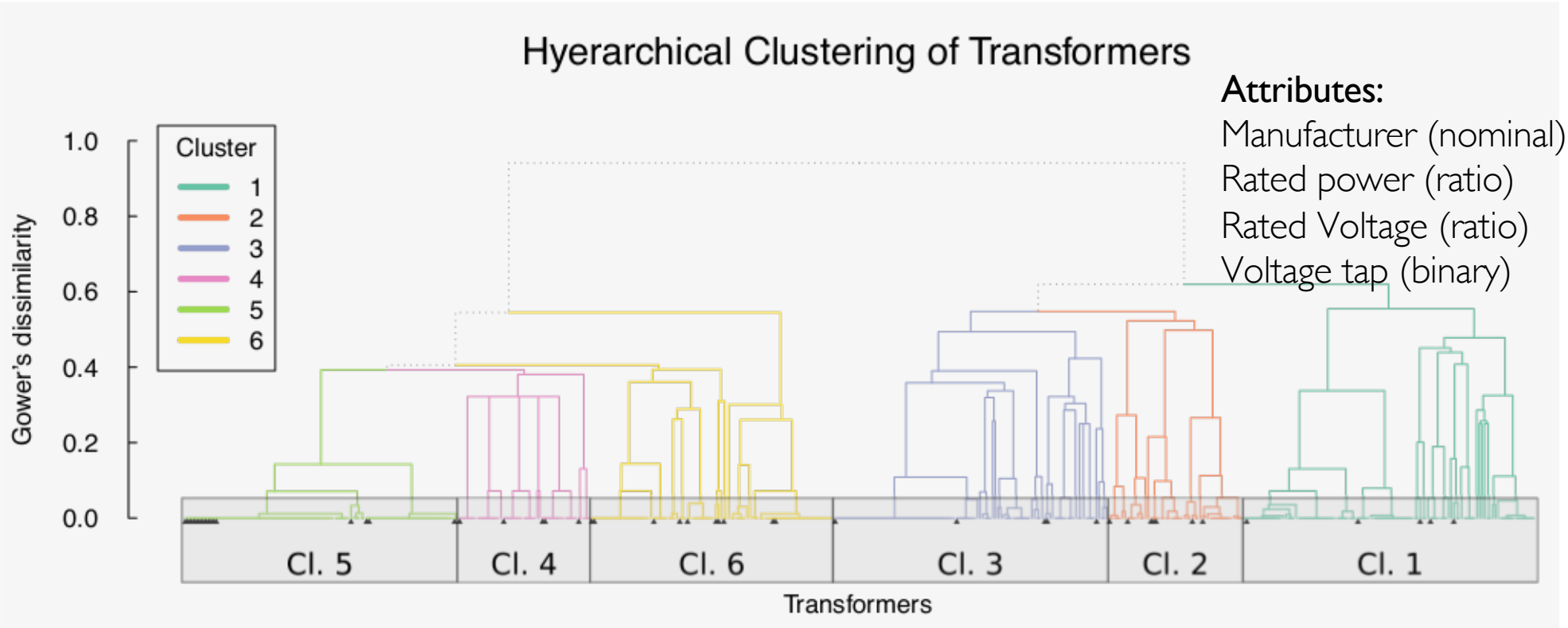


Applications of Cluster Analysis

- Clustering for understanding
  - Biological taxonomies: cladistics, evolutionary genetics, etc.
  - Stocks with similar price fluctuations
- Clustering for data summarisation
  - Reduce the size of large data sets (compression)
  - Useful, e.g., as an auxiliary step for nearest-neighbour classification (reduces computational cost)
- Clustering as a preprocessing step
  - E.g., to fit multiple regression models for relatively homogeneous subsets of data.

Example: power transformers

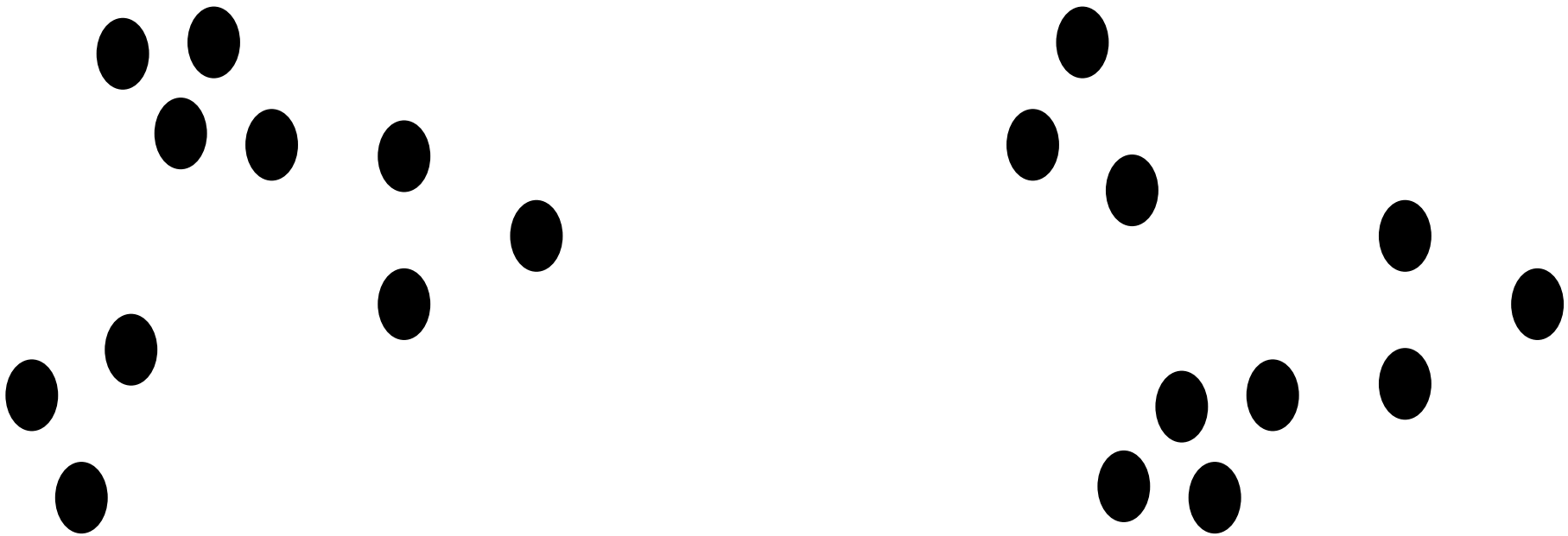
- Hierarchical clustering used as a preprocessing step for regression of expected times-to-failure, for a fleet of approx. 500 high-power transformers in Brazil.*



* https://doi.org/10.1049/iet-gtd.2015.1112

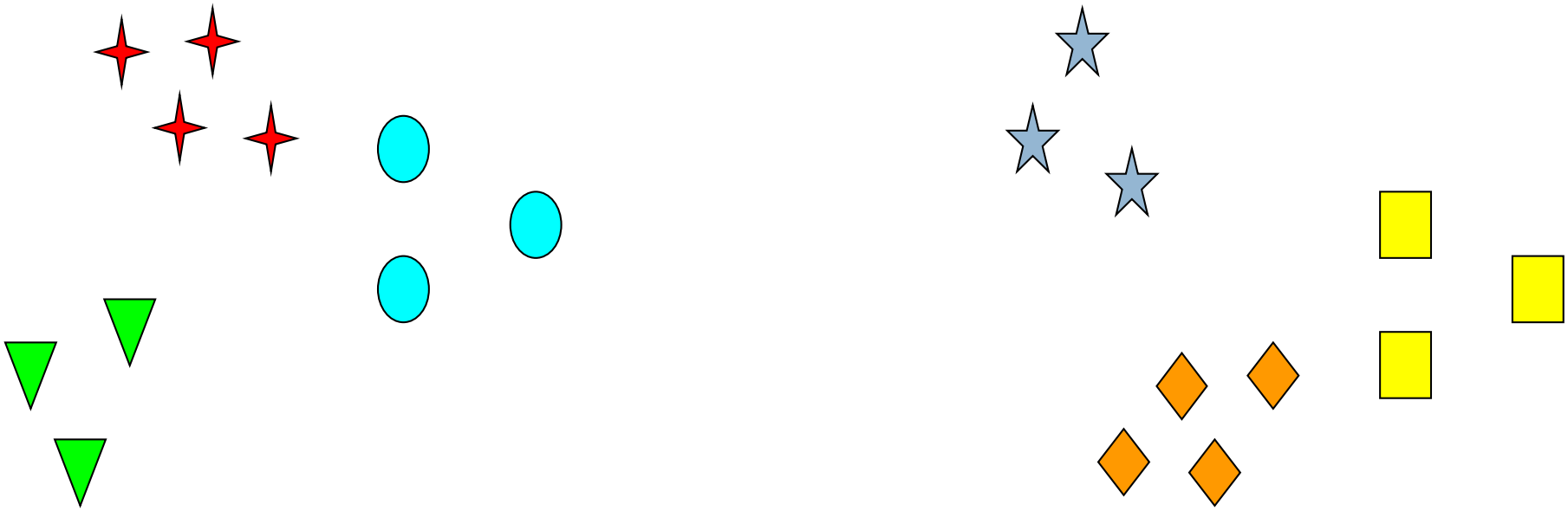
Notion of a Cluster can be Ambiguous

- How many clusters?



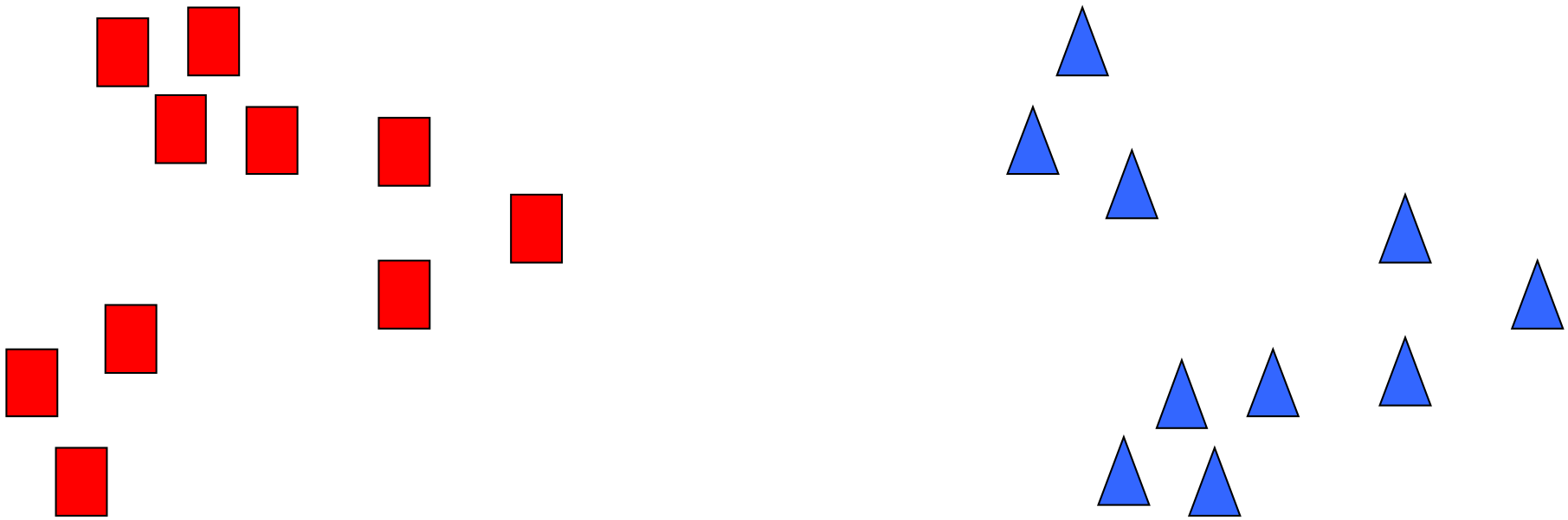
Notion of a Cluster can be Ambiguous

- How many clusters? Six?



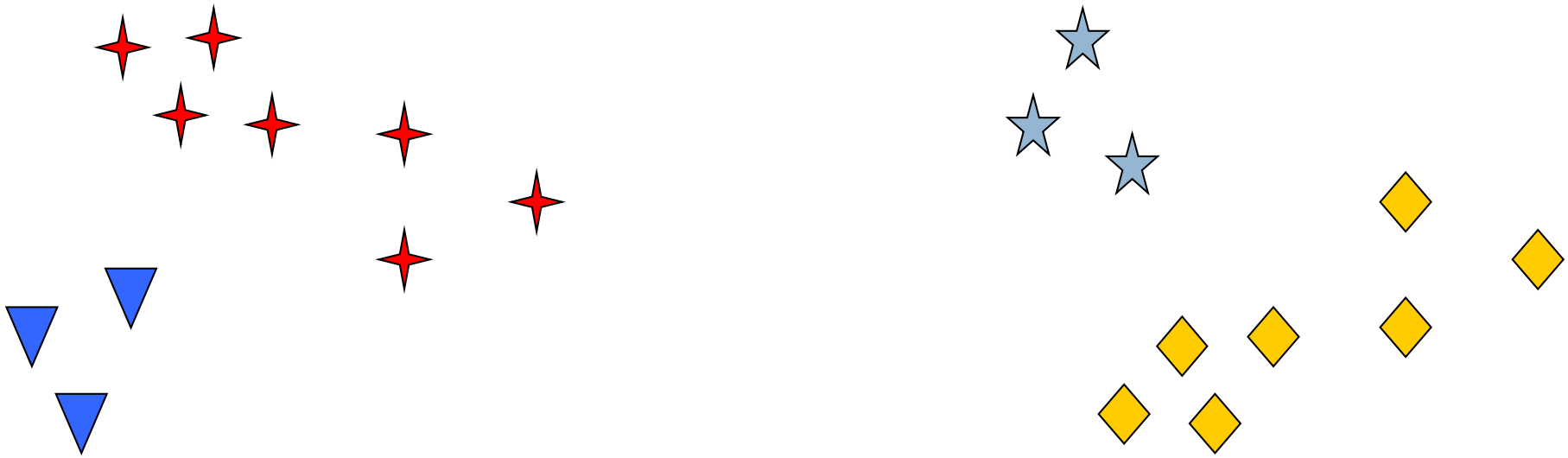
Notion of a Cluster can be Ambiguous

- How many clusters? Two?



Notion of a Cluster can be Ambiguous

- How many clusters? Or four?



Types of Clusterings

- Important distinction between
  - Partitional clustering - division of data objects into non-overlapping subsets (clusters), such that **each data object is in exactly one subset**
  - Hierarchical clustering - nested clusters, organised as a hierarchical tree.
    - clusters can have sub-clusters
    - each node in the tree (except for the leaf nodes) is the union of its children nodes, and the **root node** is the cluster containing all objects.

Other clustering concepts

- Exclusive (non-overlapping) vs. non-exclusive (overlapping)

  In non-exclusive clusterings, points may belong to multiple clusters.

- Fuzzy (or soft) vs non-fuzzy (or hard/crisp)

  In fuzzy clustering, a point belongs to every cluster with some weight (or membership value) between 0 and 1

  Weights usually must sum to 1 (often interpreted as probabilities)

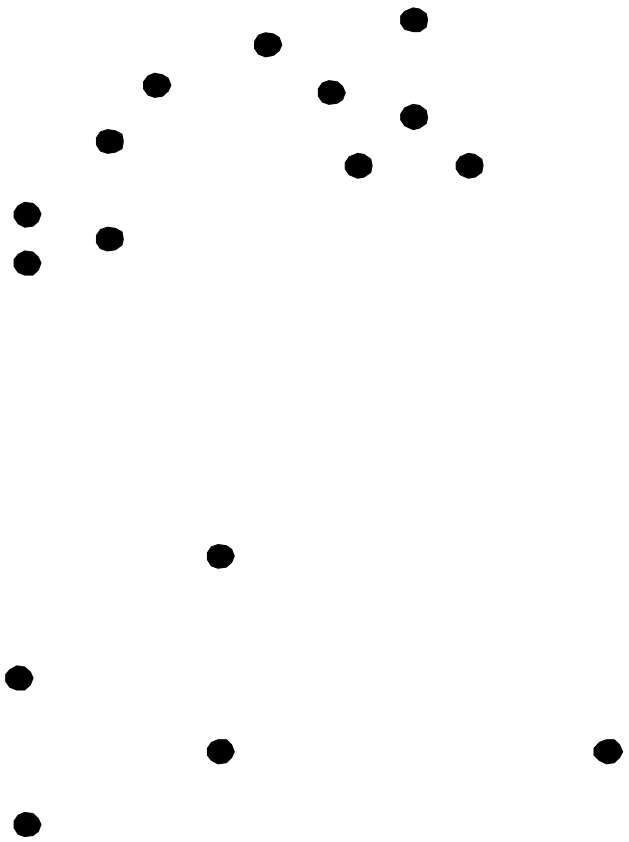
- Partial vs complete

  In some cases, we only want to cluster some of the data

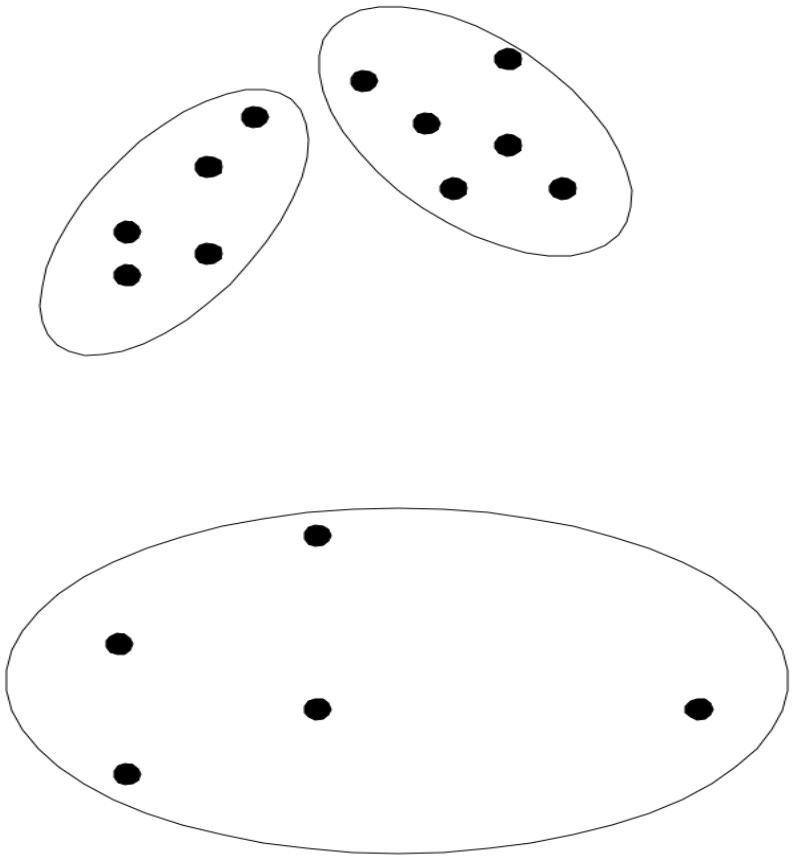
Topics

- What is Cluster Analysis
- Clustering approaches
  - Partitional: K-means
  - Density-based: DBSCAN
  - Hierarchical: Agglomerative Nesting
- Cluster Validity
  - Similarity Metric

Partitional Clustering



Original Points



A Partitional Clustering

K-means Clustering

- Partitional clustering approach
- Each cluster associated with a centroid ("center" point)
- Each point assigned to the cluster with the closest centroid
- Number of clusters (K) must be specified
- The basic algorithm is very simple

Algorithm 1 Basic K-means Algorithm.

```
Select K points as the initial centroids.
repeat
    Form K clusters by assigning all points to the closest centroid.
    Recompute the centroid of each cluster.
until The centroids don't change
```

K-means Clustering – Details

- Initial centroids are often chosen randomly
  - Clusters produced vary from one run to another
- The centroid is generally the mean of points in cluster
- Closeness measured by Euclidean distance, cosine similarity, correlation, Gower coefficient, etc.
- K-means converges (albeit not necessarily to optimality) for common similarity measures mentioned above.
- Convergence happens in the first few iterations.
  - Stopping condition can be changed, e.g., to '***until relatively few points change clusters***'

Evaluating K-means Clusters

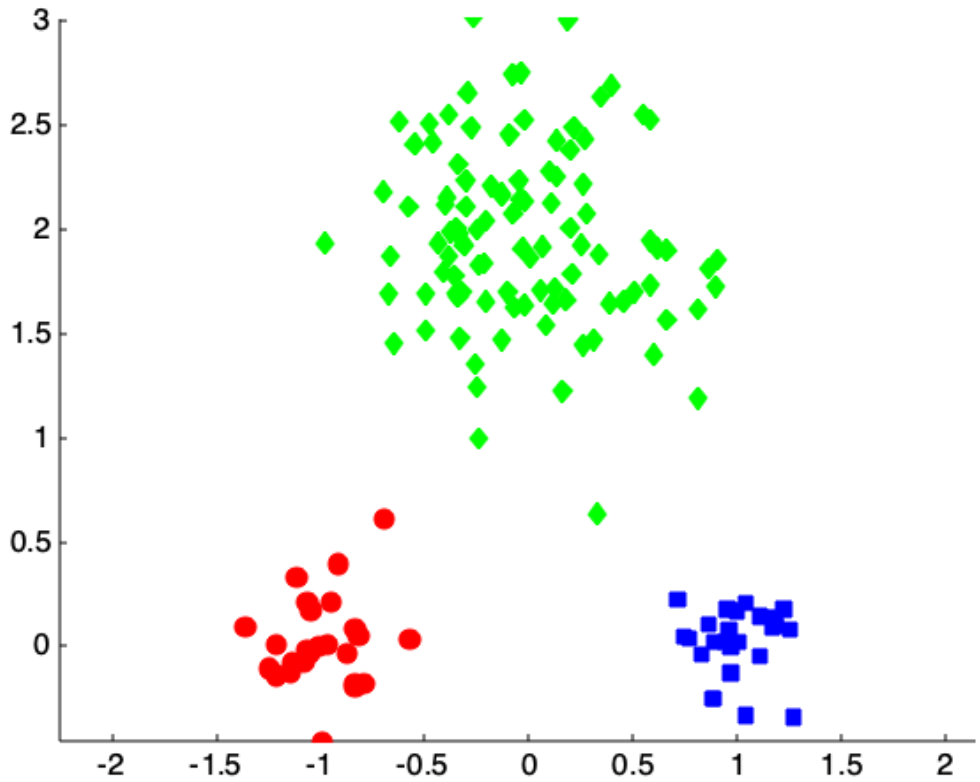
- K-means is essentially trying to minimise a cost function that is the sum of within-cluster dissimilarities.
- Most common measure: total within-clusters Sum of Squared Errors (SSE):

$$SS_E = \sum_{i=1}^{K} \sum_{\forall x_j \in C_i} dist^2(x_j, m_i)$$

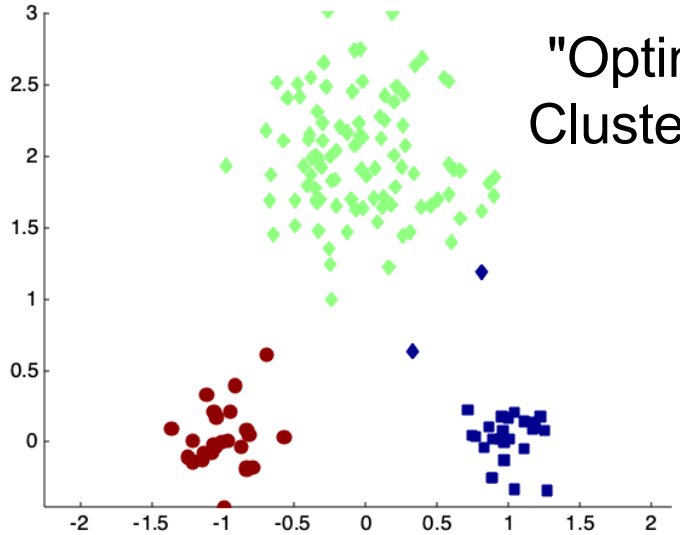
where m_i is the centroid associated to cluster C_i.

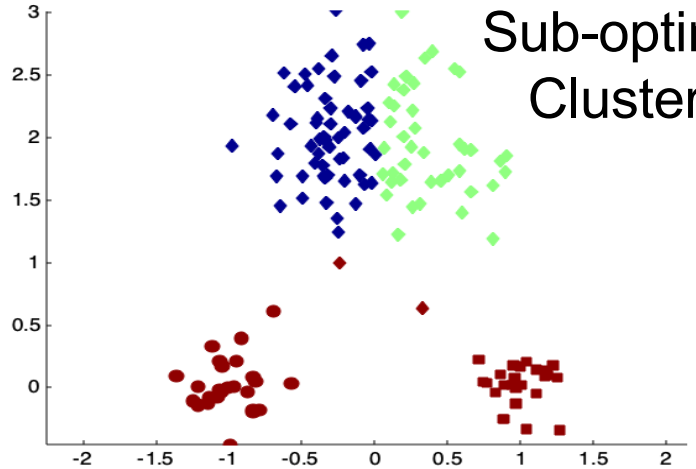
  - Given two clusterings, we can select the one with the smallest SSE.
  - One easy way to reduce SSE is to increase K, the number of clusters (where can it lead in the extreme case?)

Importance of initial centroids



Simulated Points





Common initialization approaches

- Perform multiple runs under different random initial conditions, then use the one with the smallest error.

- Select initial centroids by some non-random method, e.g., using the most diverse – mutually distant – k-subset of the data as the initial cluster centers (**k-means++** algorithm).

More on centroids

- The centroid depends on the dissimilarity function used – commonly Euclidean distance for relatively low-dimensional problems with only numeric attributes, other functions used in different cases

  e.g., cosine dissimilarity, Jaccard's coefficient, Gower's dissimilarity

- Variations:

  k-medoids (use an actual point of the cluster as center point instead of the mean)

  K-centers (changes goal to minimisation of the maximum within-clusters dissimilarity).

Pre-processing and Post-processing

- Pre-processing
  - Normalise the data
  - Eliminate outliers

- Post-processing
  - Eliminate small clusters (that may represent outliers)
  - Split 'loose' clusters, i.e., clusters with relatively high SSE
  - Merge clusters that are 'close' and that have relatively low SSE

Selecting K

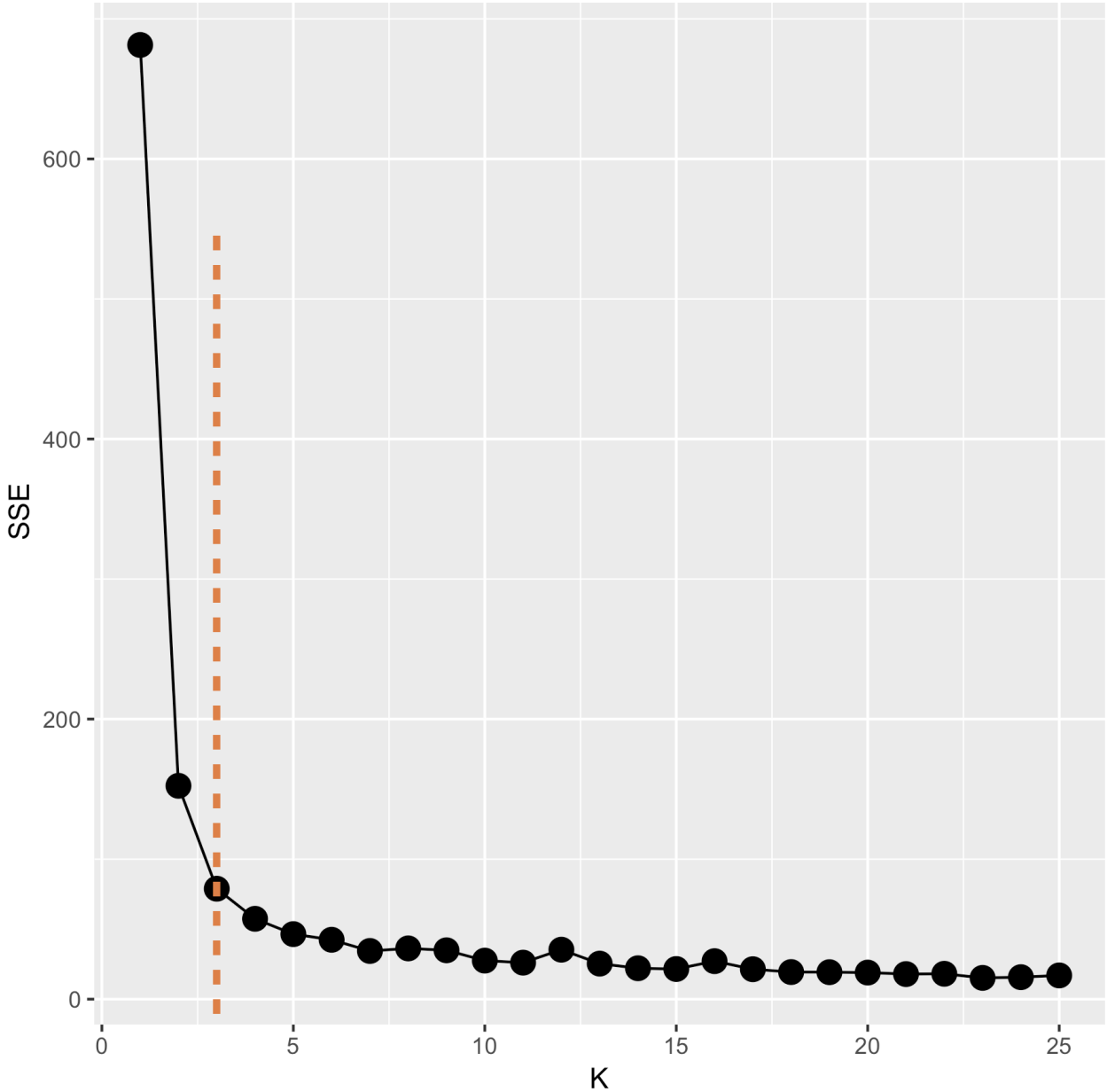
- Sometimes it is not easy to select K in advance (particularly for high-dimensional data).
- Several methods available:
  - Elbow method
  - Gap Statistic
  - Silhouette Method
  - Sum of Squares Method
  - Association with hierarchical methods
  - …

For more, see *Towards Data Science:* 10 Tips for Choosing the Optimal Number of Clusters - https://goo.gl/e1wdGh

Selecting K: elbow method

- Very simple approach:
  - Iteratively increase number of clusters
  - Observe total SSE - SSE gains tend to flatten after a certain point
  - Select K with good tradeoff between model complexity (number of clusters) and SSE

Elbow method: example



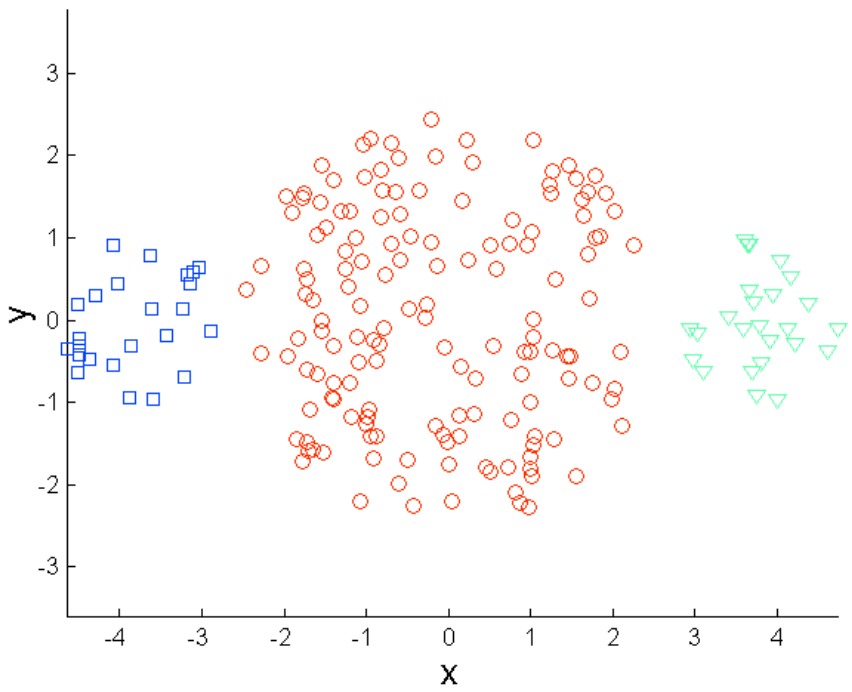
Elbow method example using the full ***iris*** dataset as a clustering example (i.e., without informing the classes).

Suggests 3 (or maybe 4?) clusters.

Limitations of K-means

- K-means has problems when
  - clusters are of differing
    - Sizes
    - Densities
    - Non-globular shapes
  - data contains outliers

Limitations of K-means: Different Sizes



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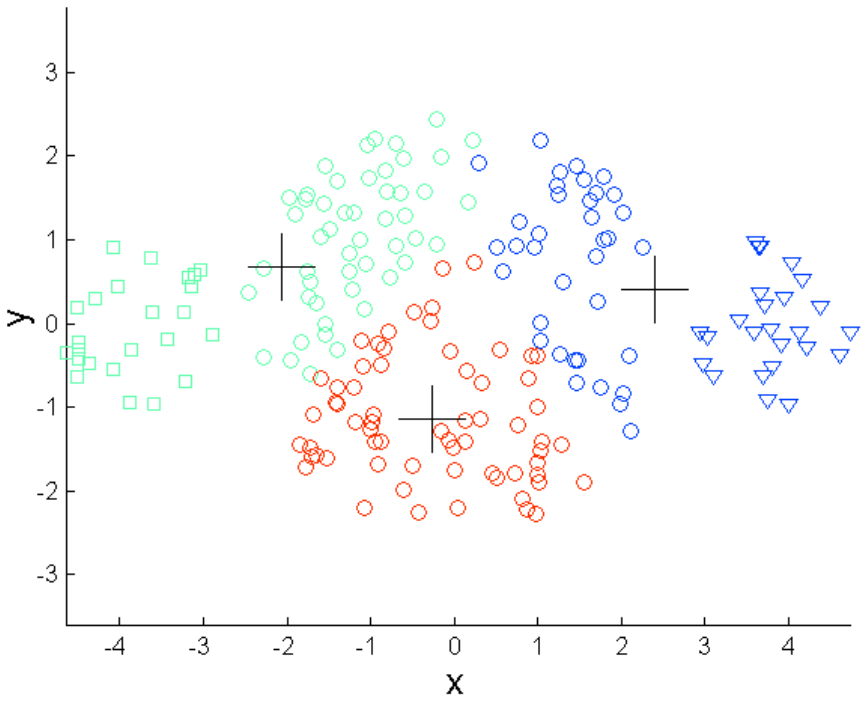
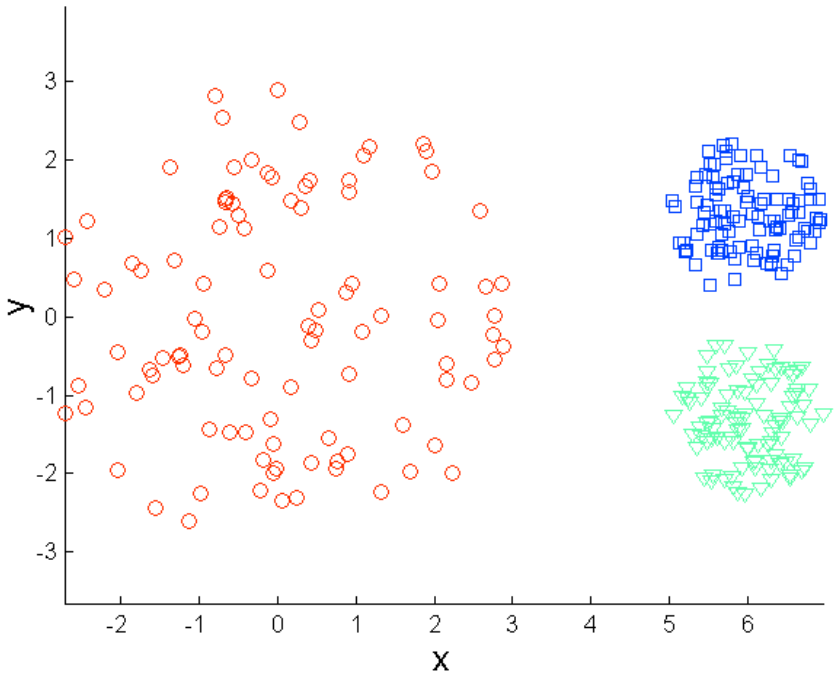
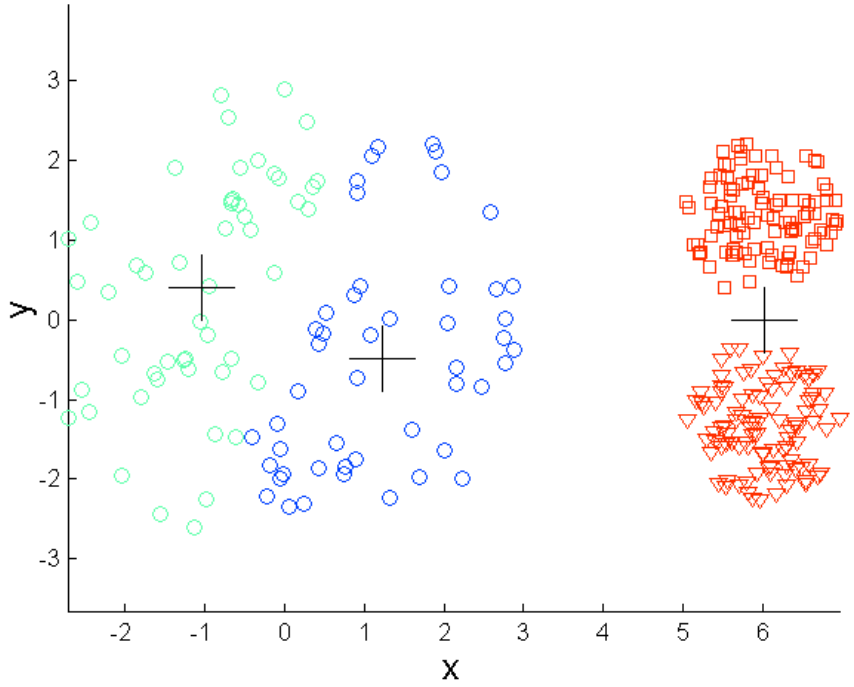


K-means (K = 3)

Limitations of K-means: Different Density

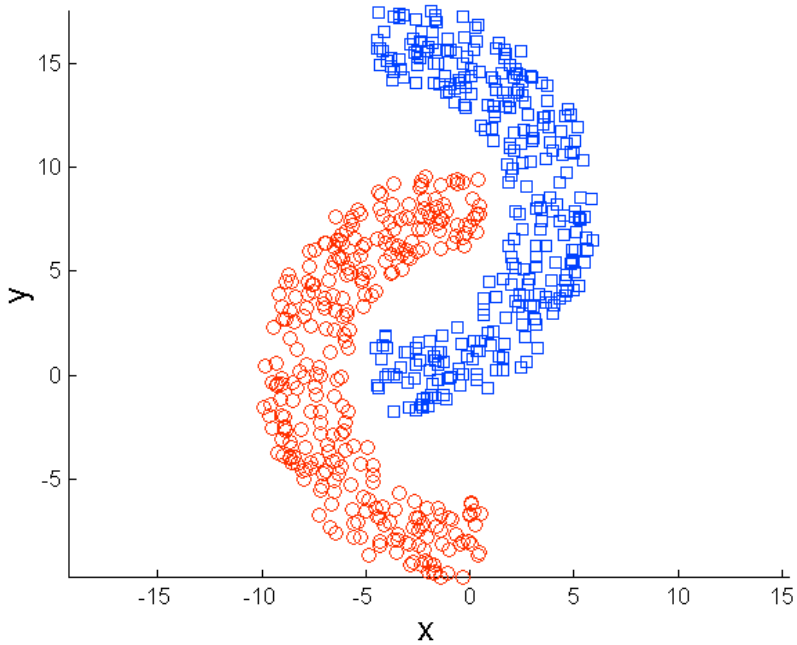


Simulated Points



K-means (K = 3)

Limitations of K-means: Non-globular Shapes



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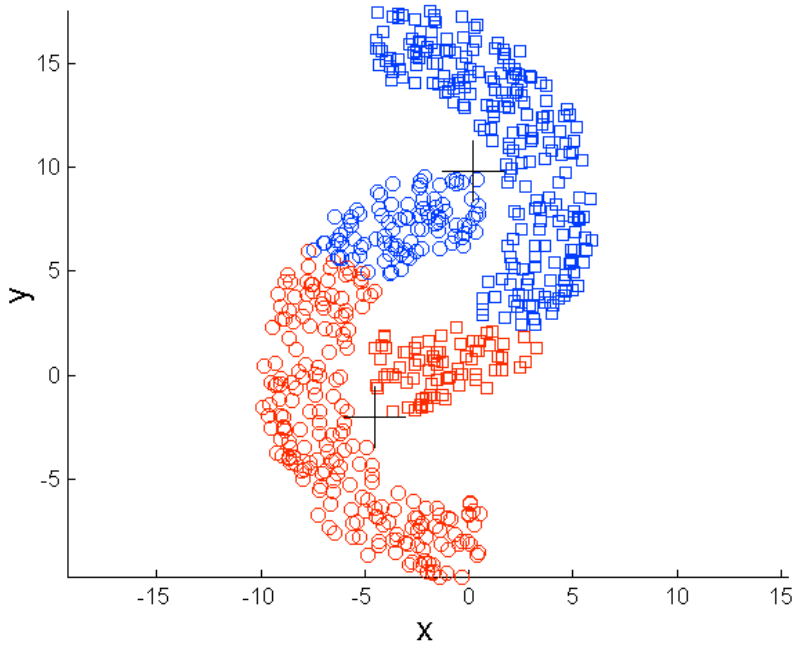
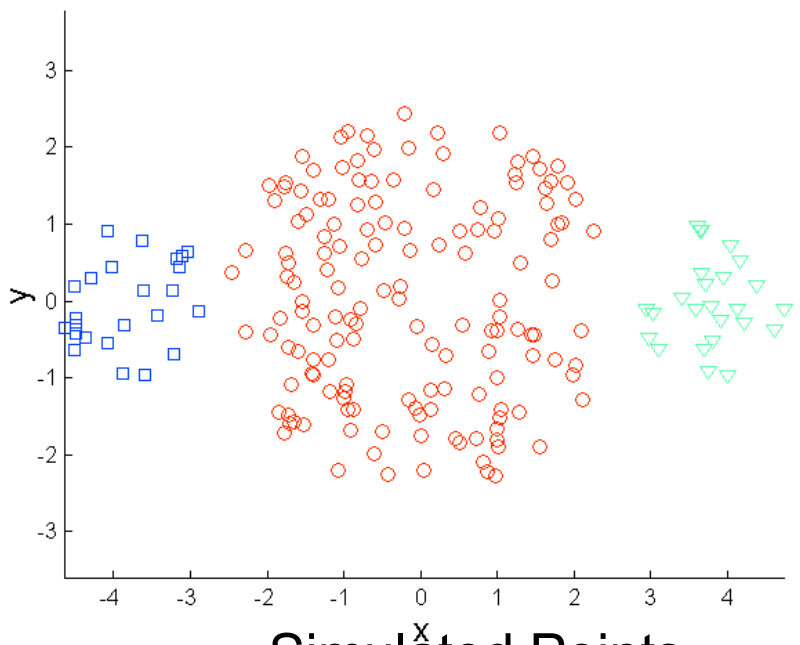


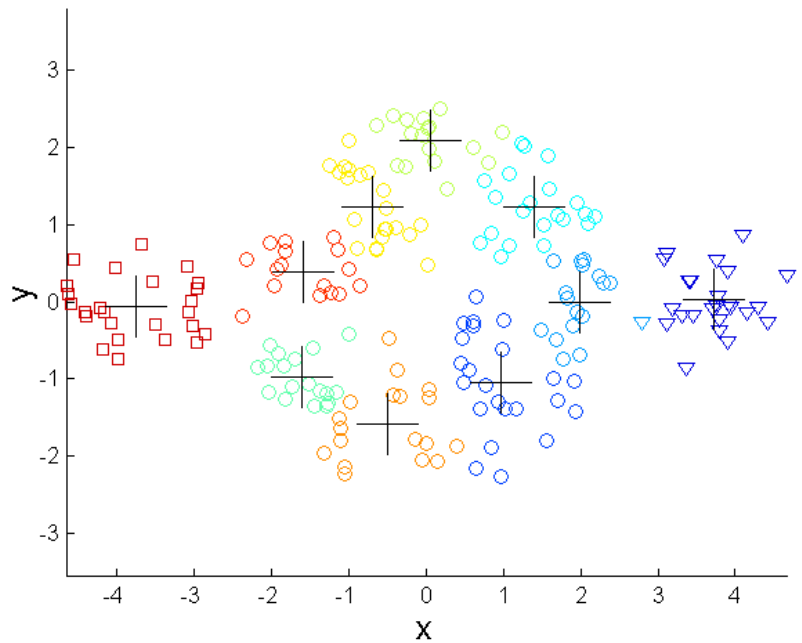
K-means (K = 2)

Overcoming K-means Limitations

- A common solution:
  - Find **many** clusters
  - Merge together

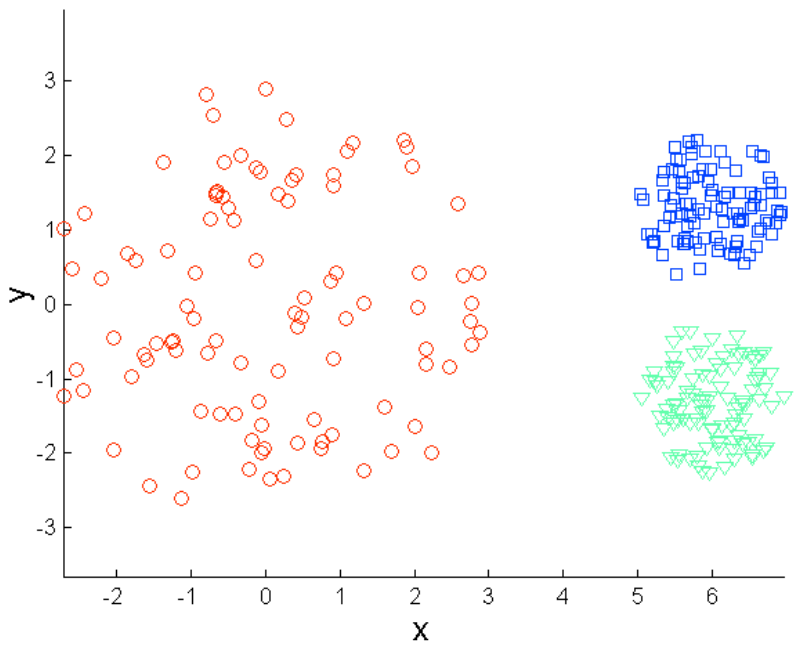


Simulated Points

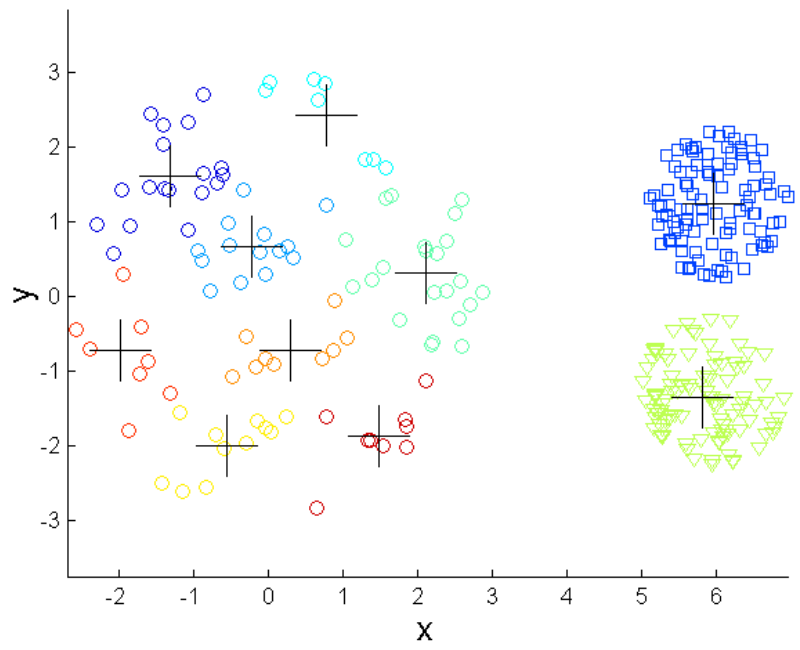


K-means (large K)

Overcoming K-means Limitations

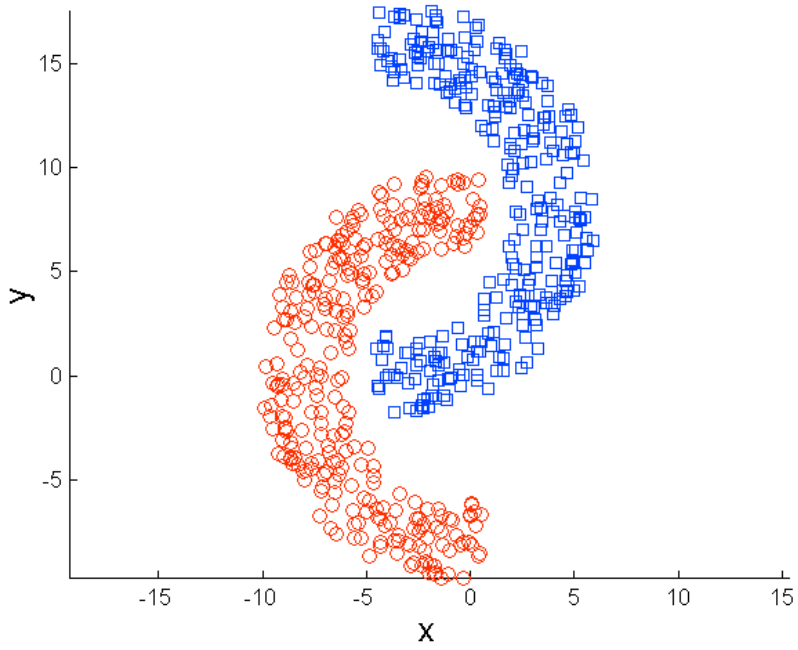


Simulated Points



K-means (large K)

Overcoming K-means Limitations



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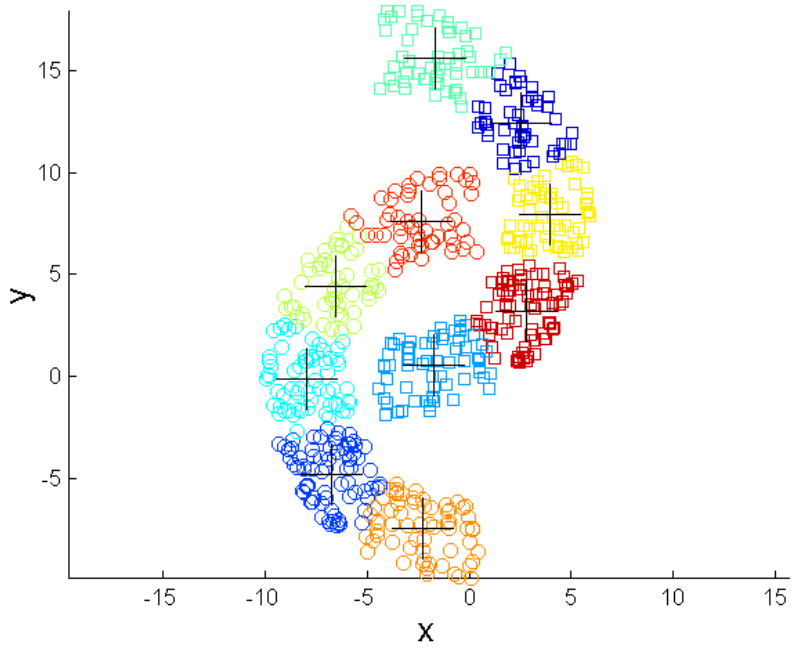

K-means (large K)

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Density-based clustering: DBSCAN

- DBSCAN is a Density-Based Clustering algorithm

  partition points into **dense** regions separated by not-so-dense regions.

- Important Questions:

  How do we measure density?

  What is a dense region?

- In the case of DBSCAN:

  Density at point *p*: number of points within a ball of radius *Eps.*

  Dense Region: a ball of radius *Eps* that contains at least *MinPts* points.

For a good short intro to DBSCAN, check http://dx.doi.org/10.18637/jss.v091.i01 Section 2.1

Types of points in DBSCAN

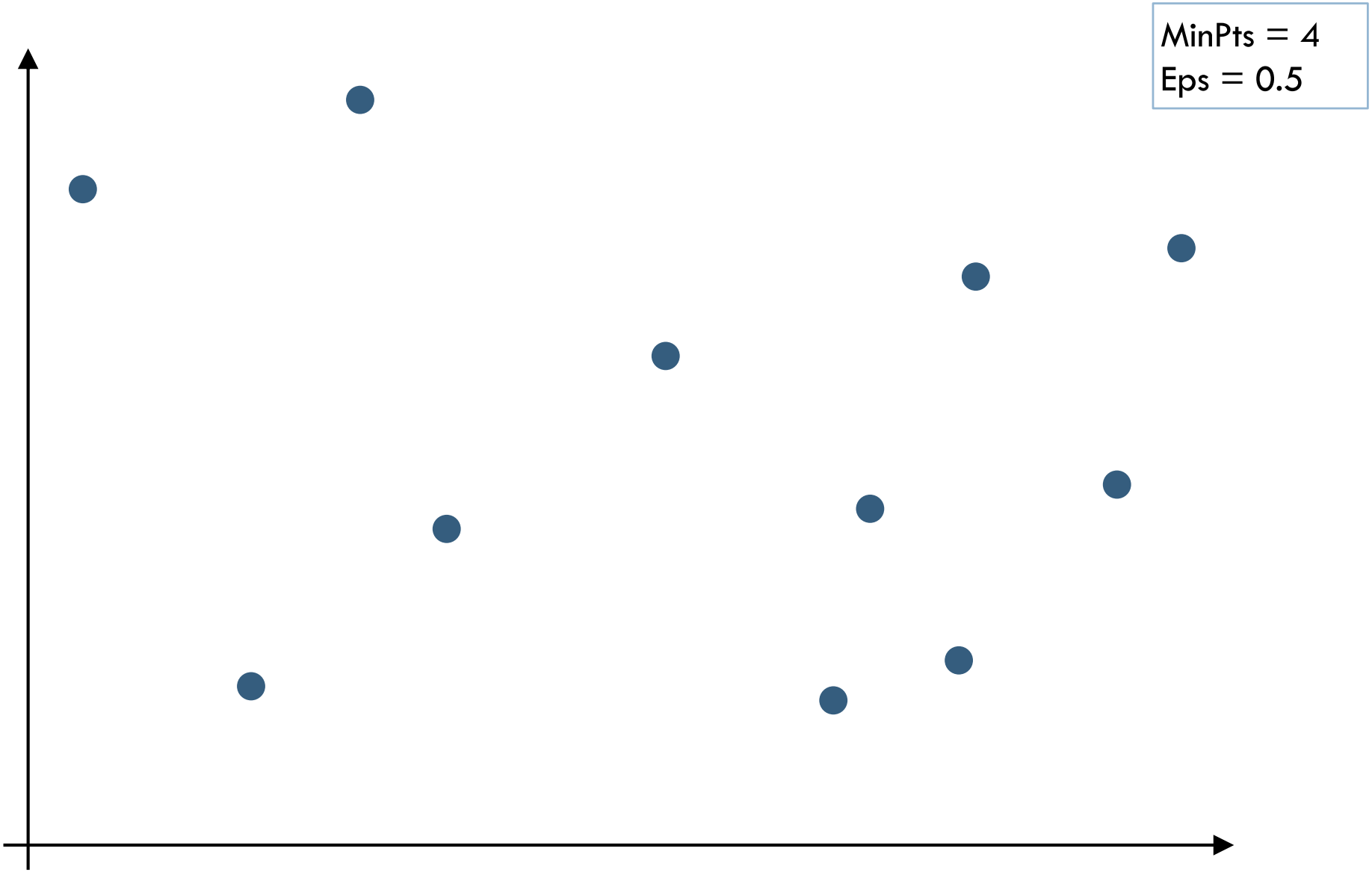
- In DBSCAN a point can be characterised as:

  A core point, if it has more than *MinPts* within a distance *Eps*. **Core points** belong in a dense region and are in the interior of a cluster.

  A border point, if it has fewer than *MinPts* points within a distance *Eps*, but is located in the *Eps*-neighbourhood of at least one core point.

  A noise point, if it is neither a core point nor a border point.

Types of points in DBSCAN



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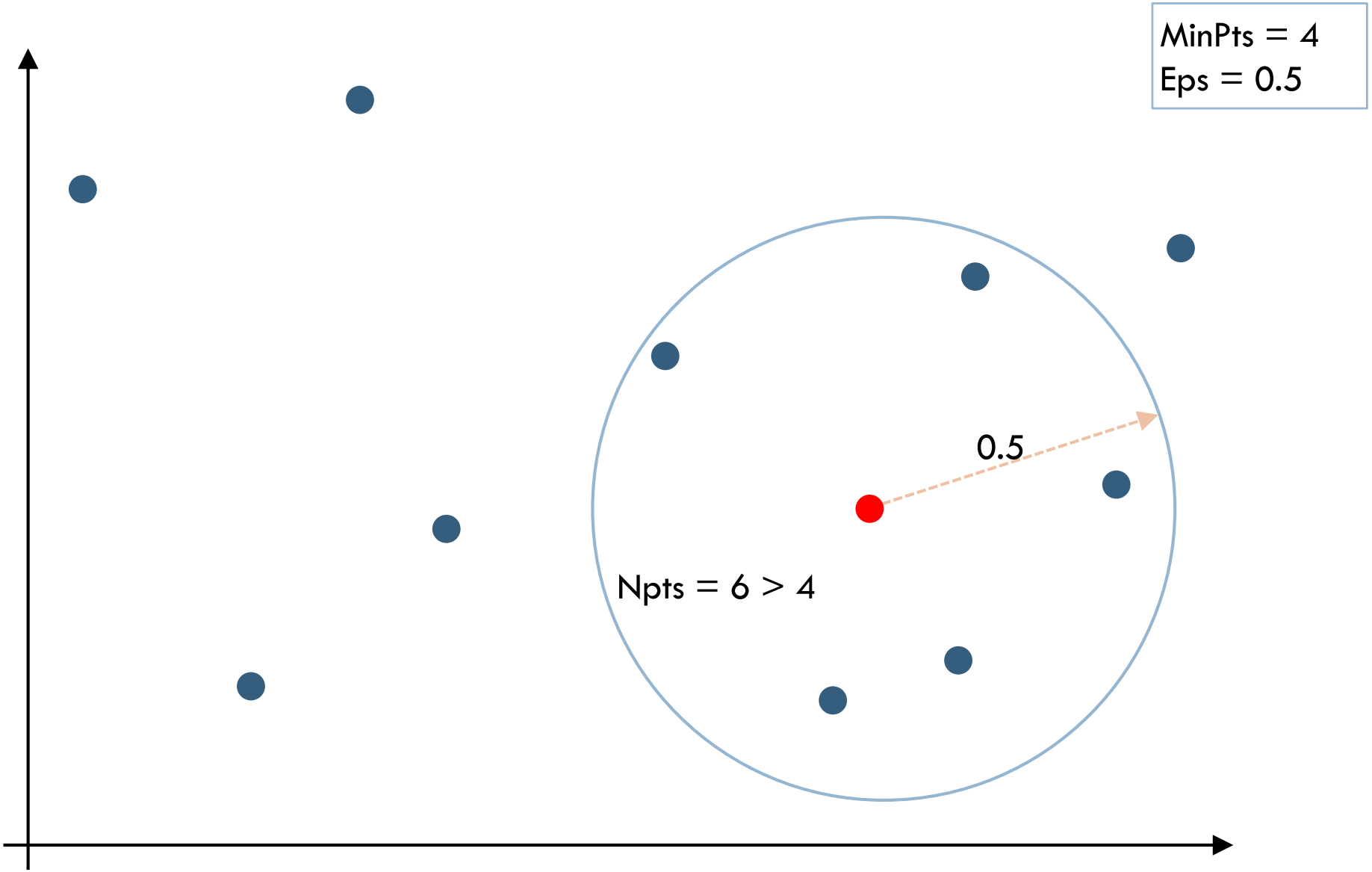

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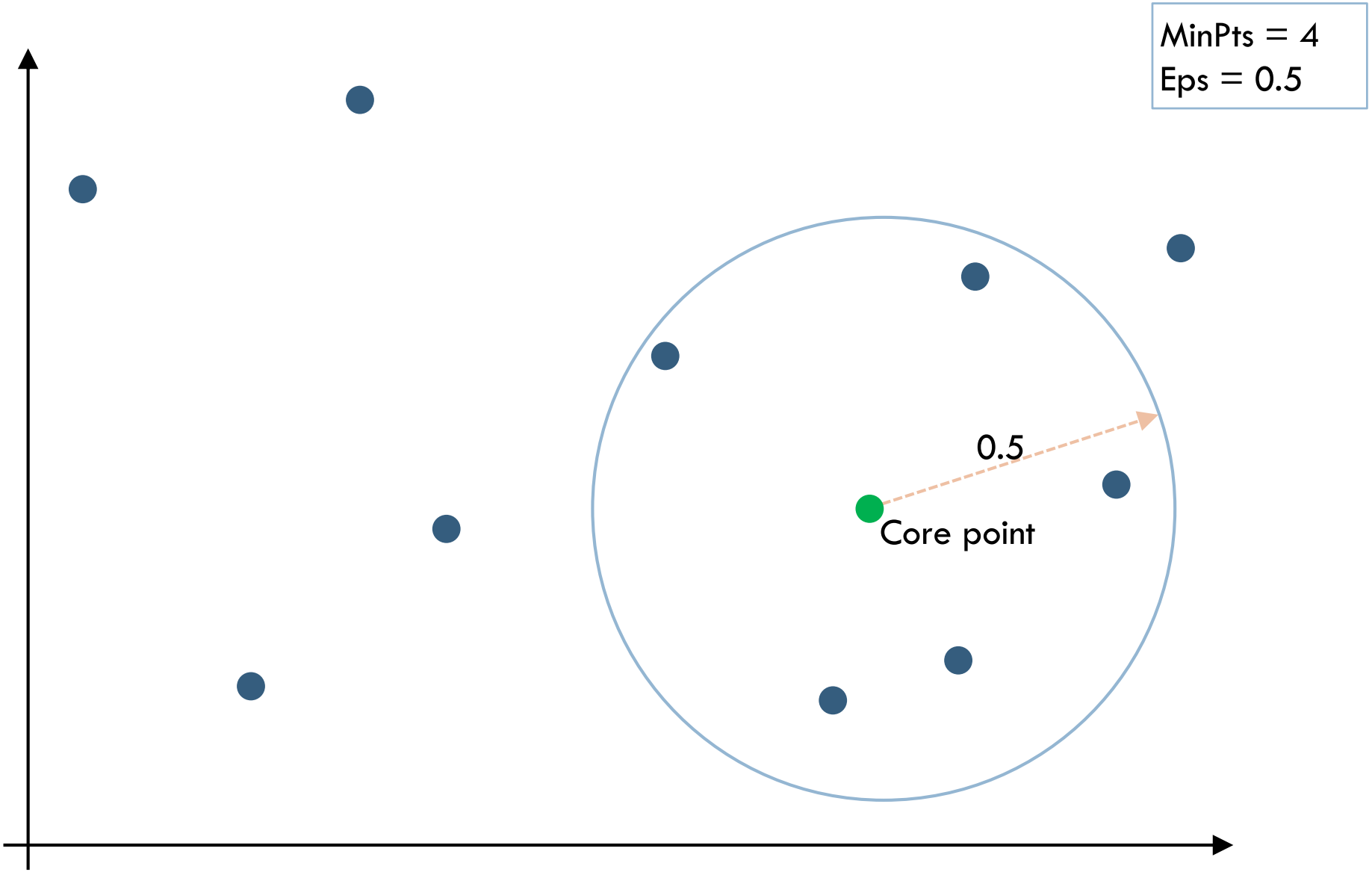

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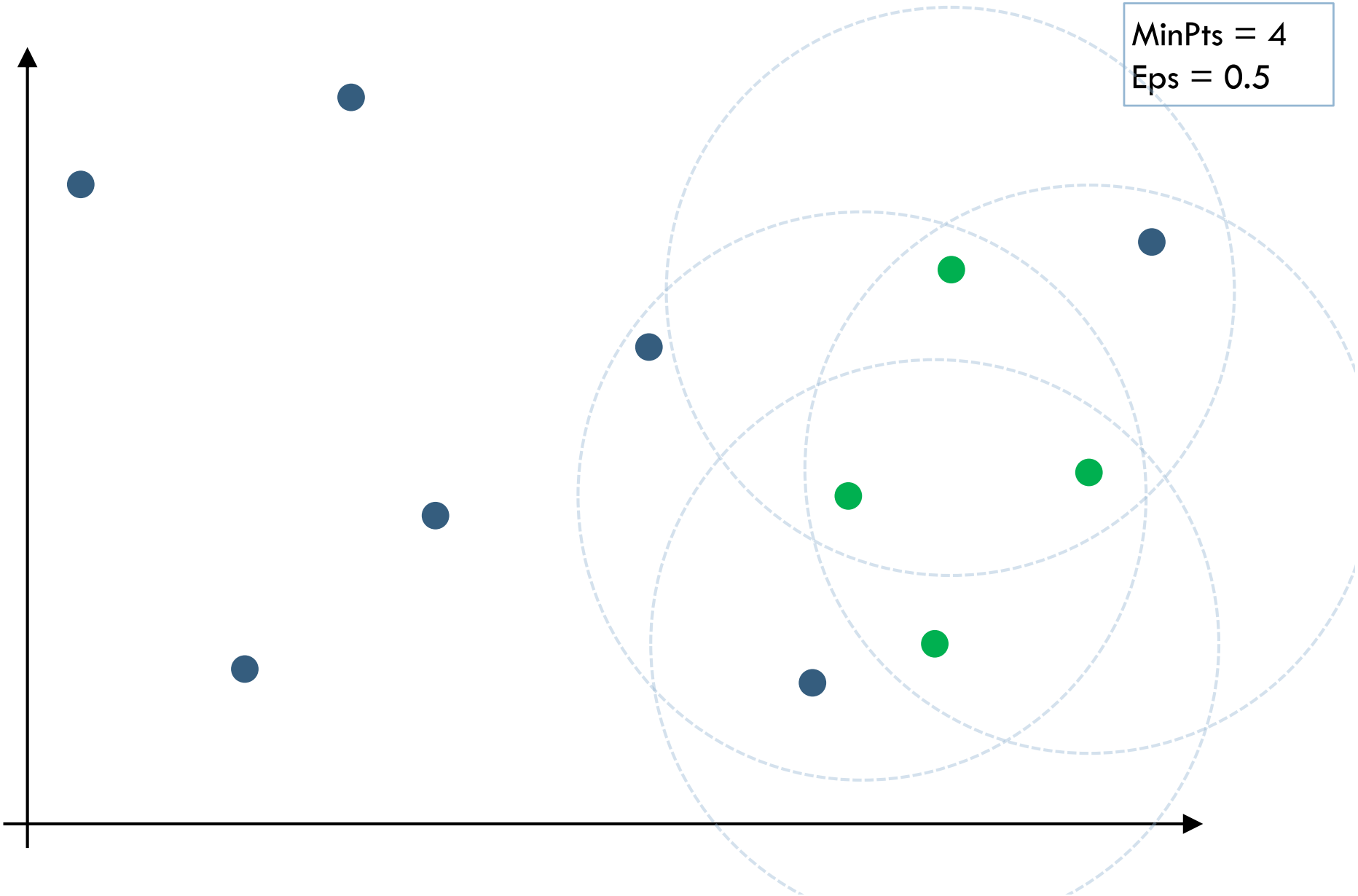

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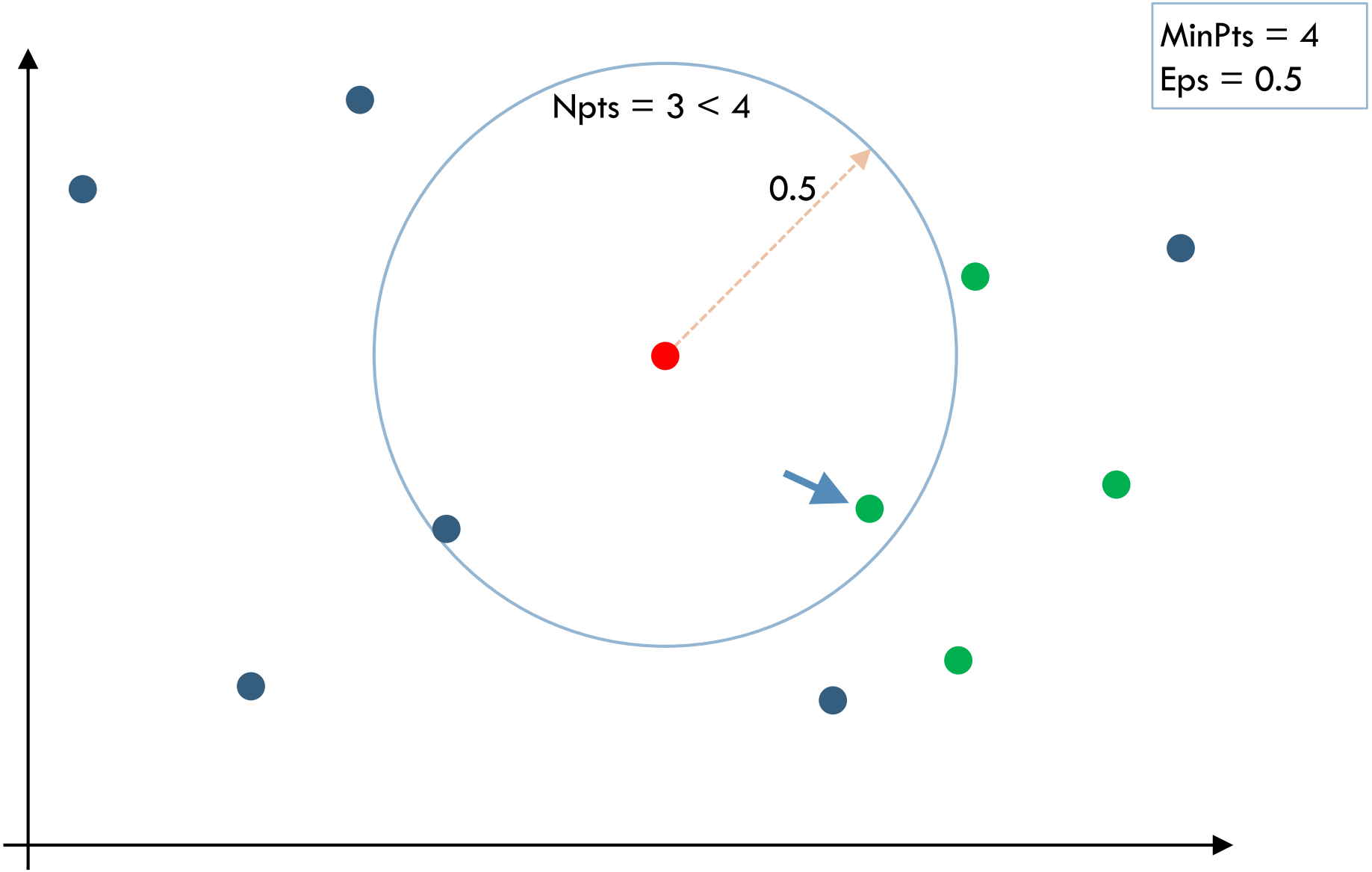

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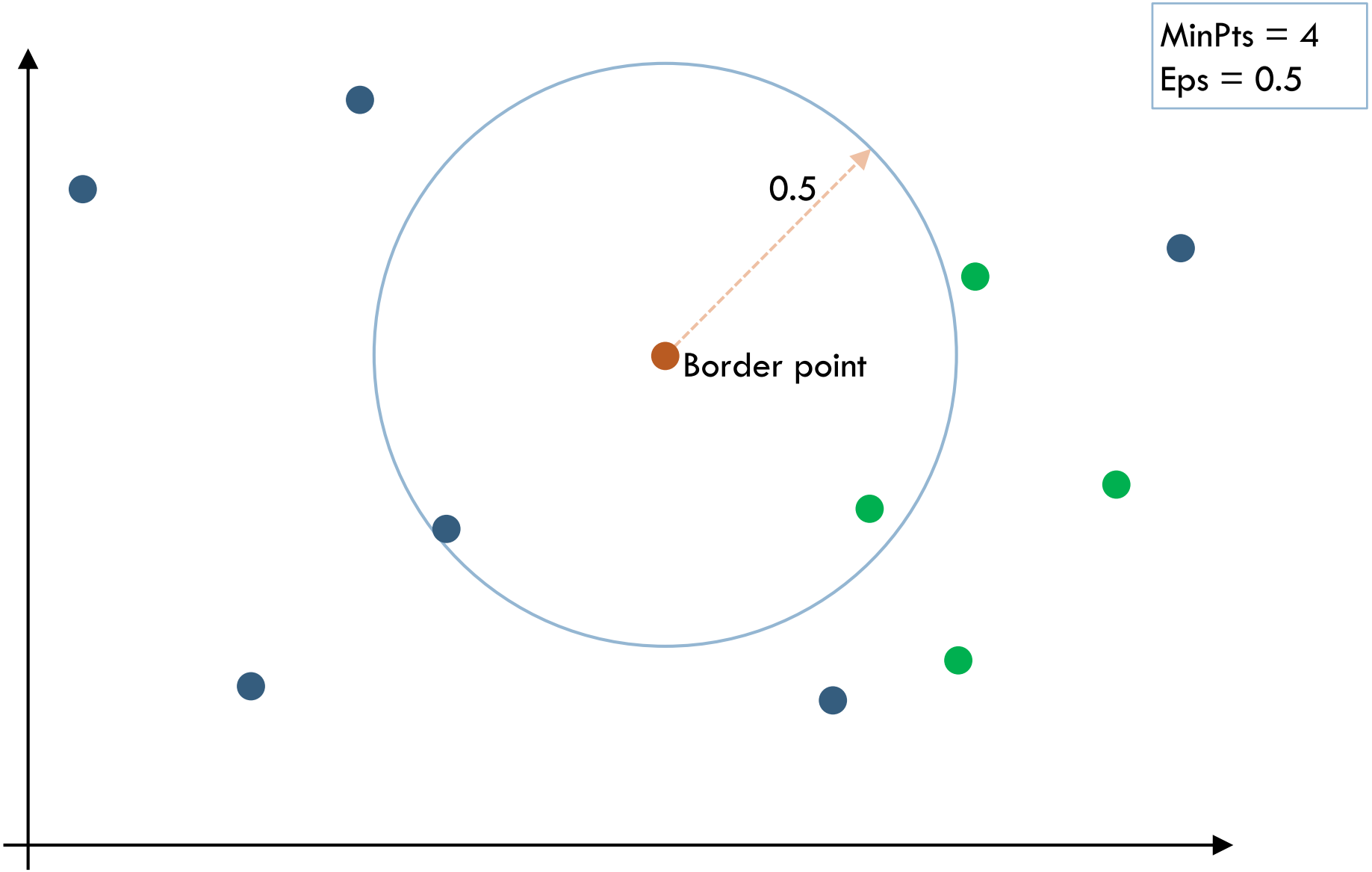

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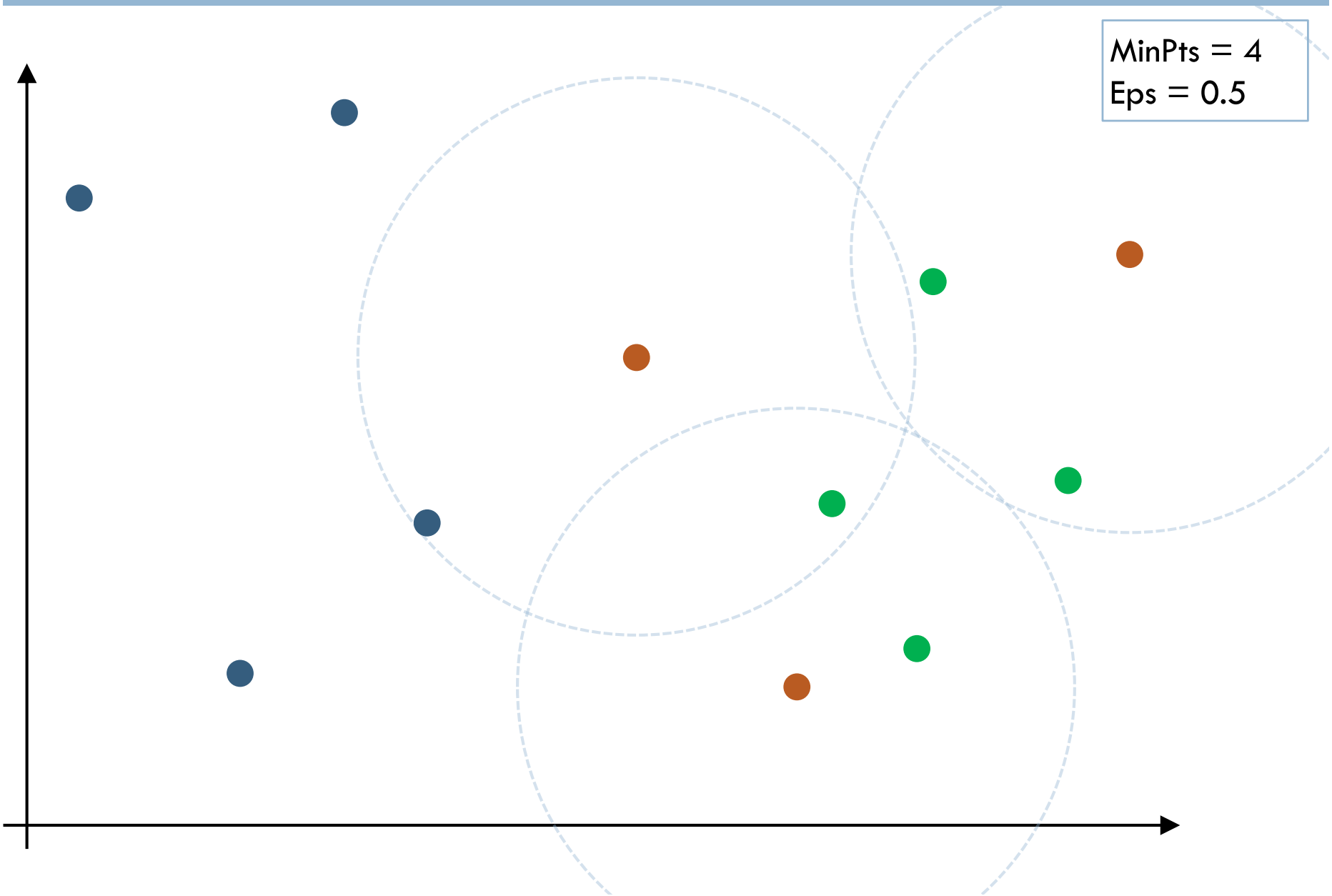

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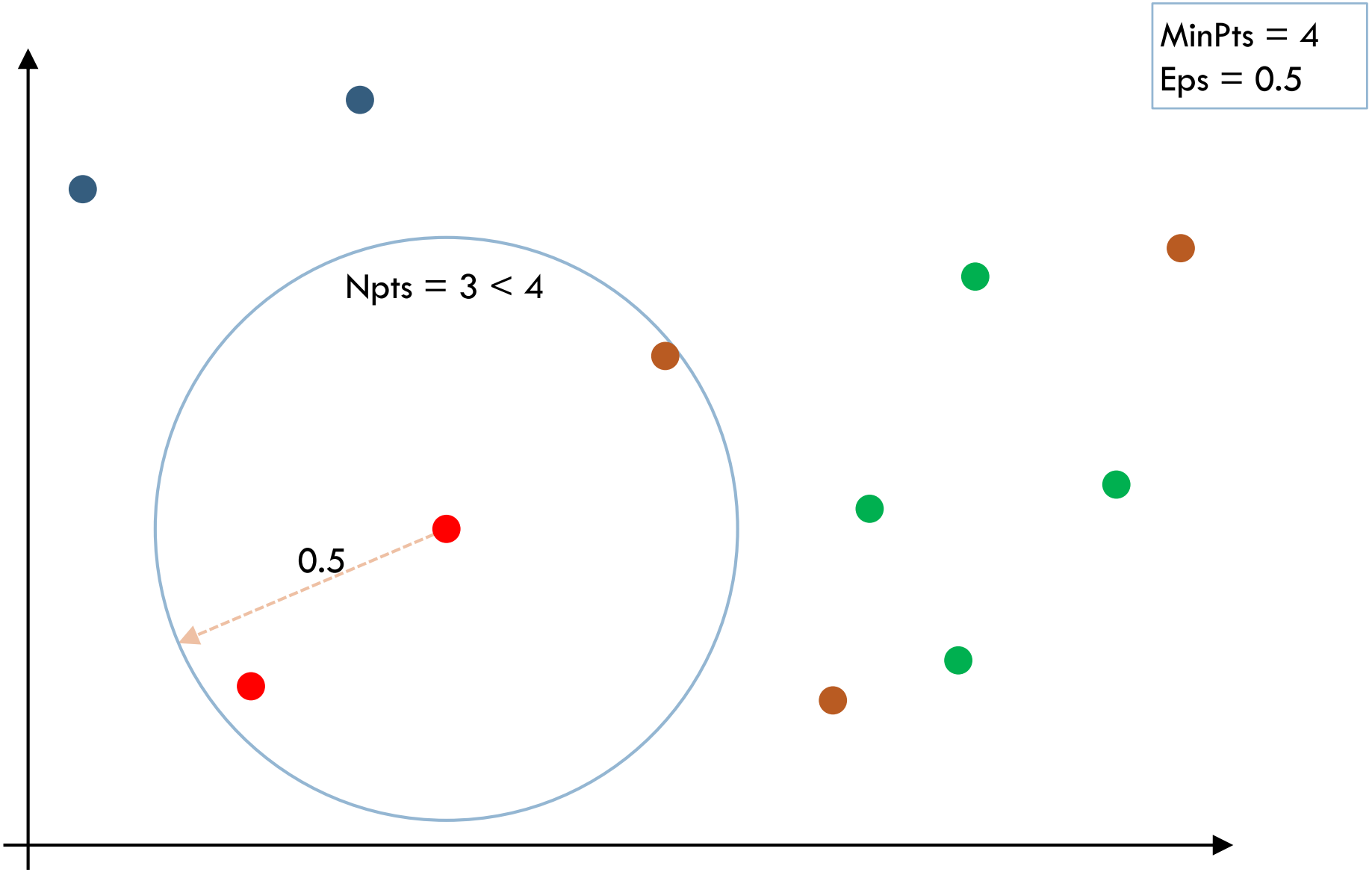

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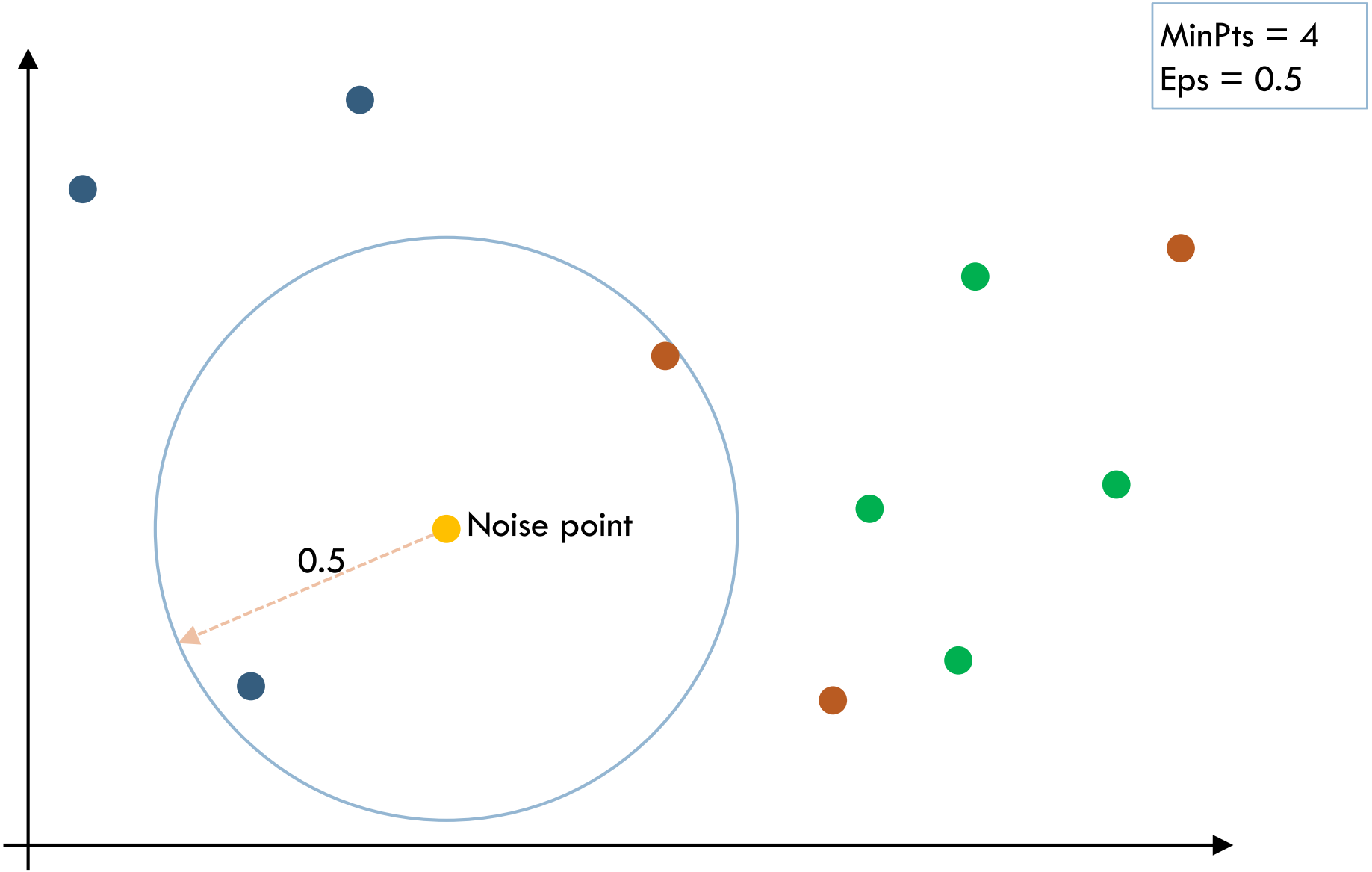

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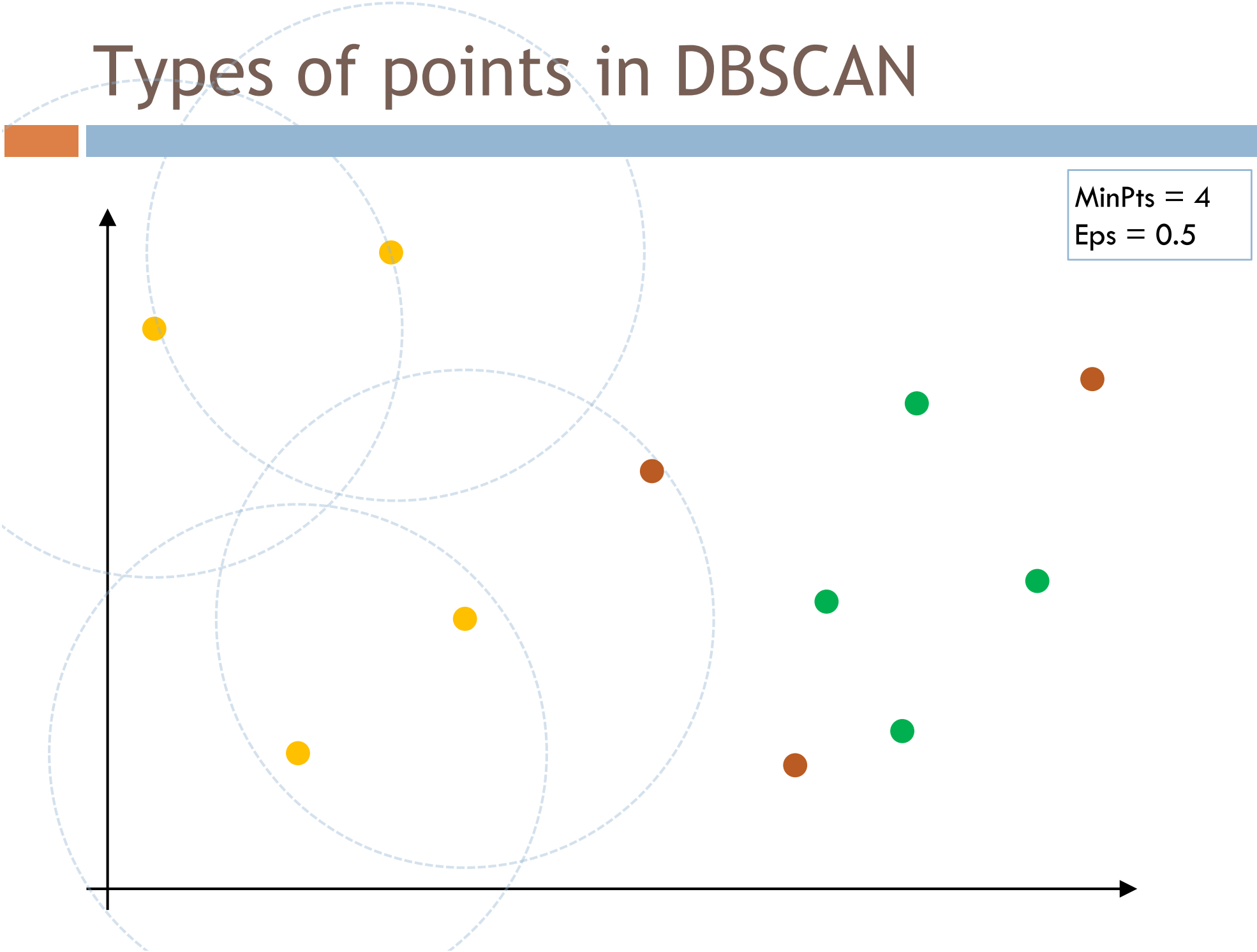
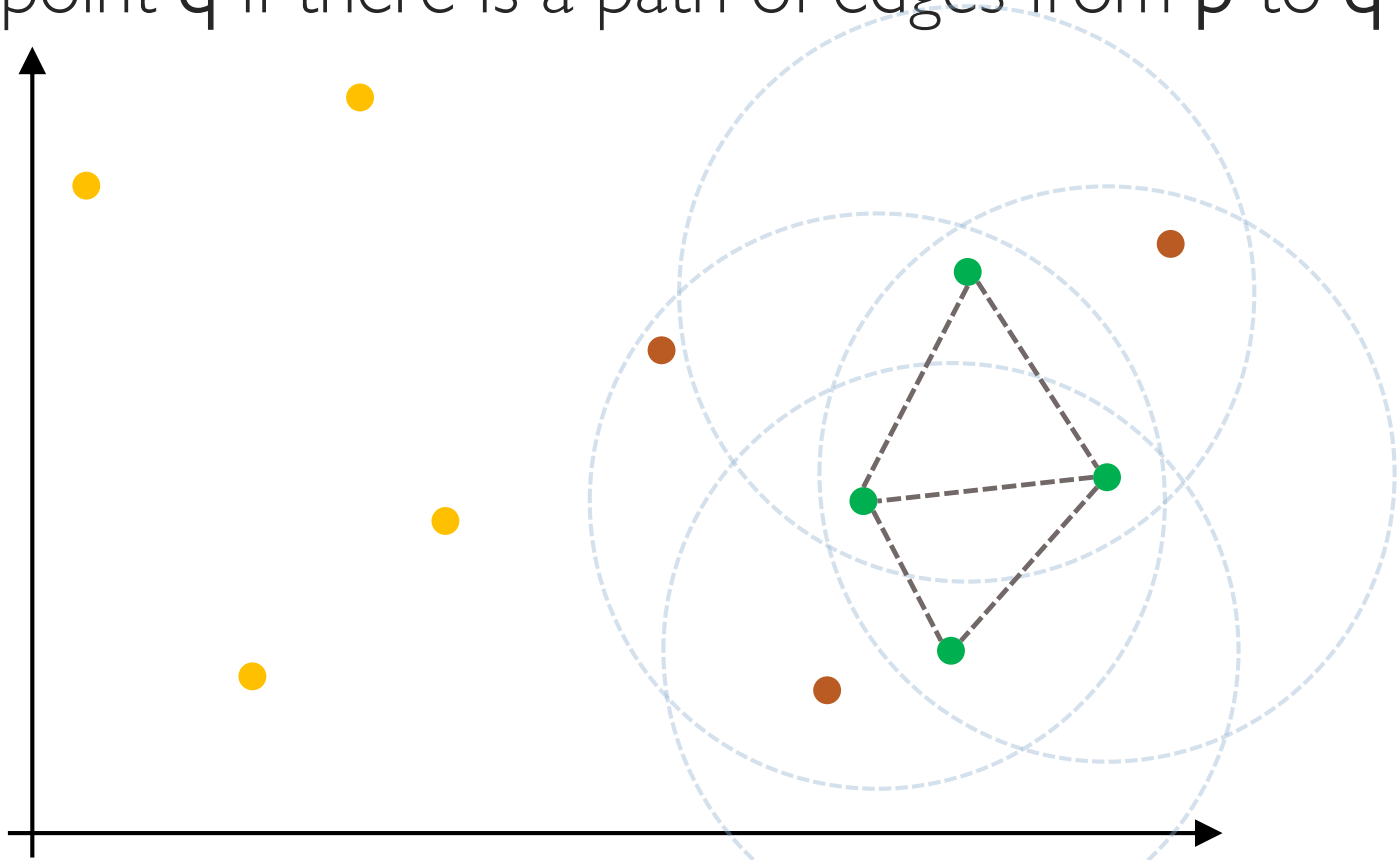


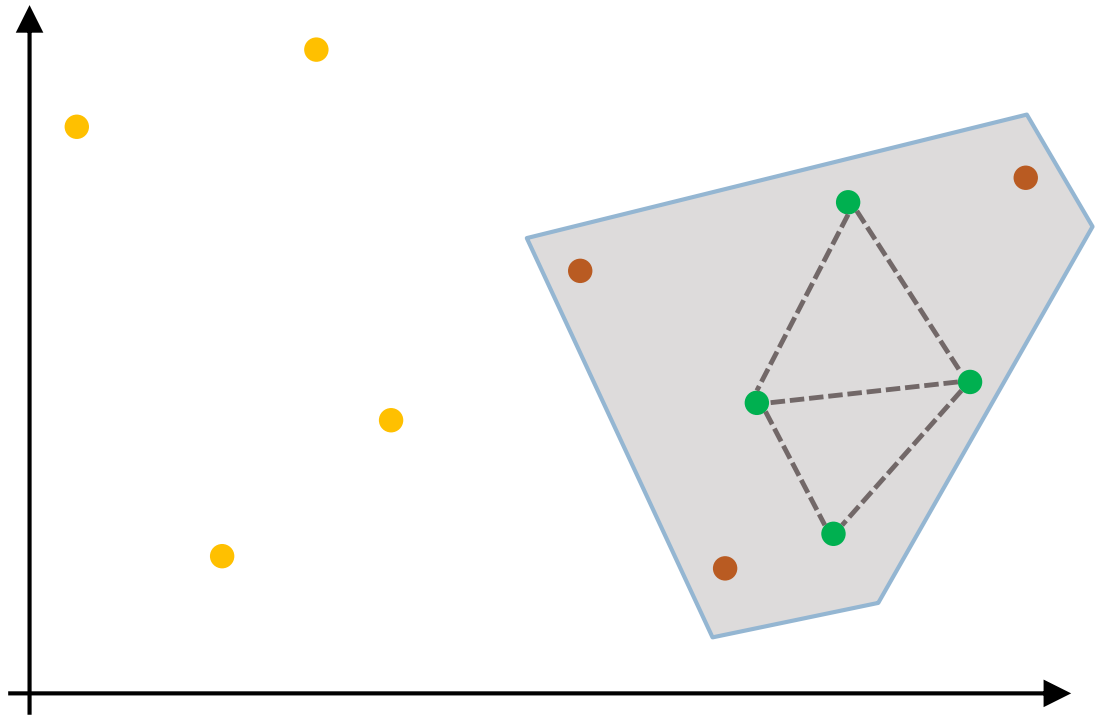
Density-connected points

- *Density edge*: we place an edge between two core points q and p if they are within distance *Eps.*
- *Density-connected points*: A point p is *density-connected* to a point q if there is a path of edges from p to q



DBSCAN algorithm

- Label points as core, border or noise.
- Eliminate noise points.
- Put an edge between all core points that are within *Eps* of each other
- Make each group of connected core points into a separate cluster.
- Assign each border point to one of the clusters of its associated core points.



DBSCAN: determining *Eps* and *MinPts*

- A simple heuristic:

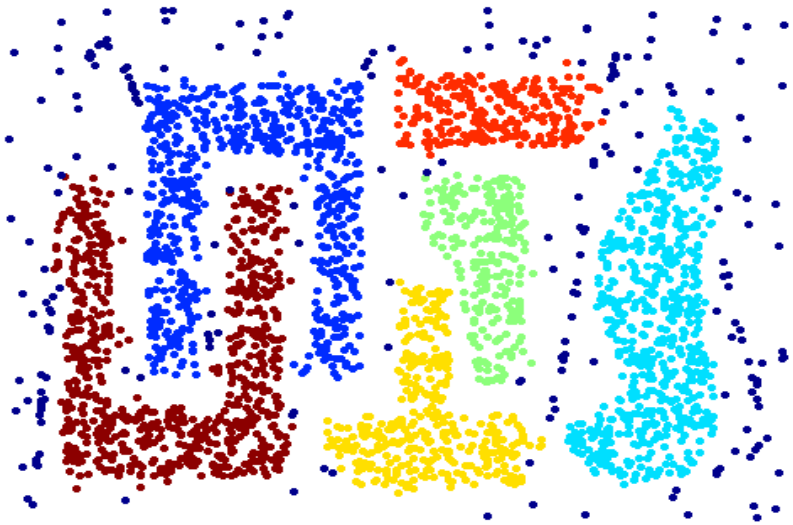
  Compute the distance between all points and their k-th nearest neighbour (for some small k, e.g., 4 or 5). We'll call this distance d_k.

  Sort the d_k values in increasing order and then plot the sorted values. We expect to see a sharp change at the value of d_k that corresponds to a suitable value of *Eps*.

  If we select this d_k value as the *Eps* parameter and take the value of k as the *MinPts* parameter, then points for which $d_k < Eps$ will be labeled as core points, while other points will be labeled as noise or border points.

DBSCAN

Original Points

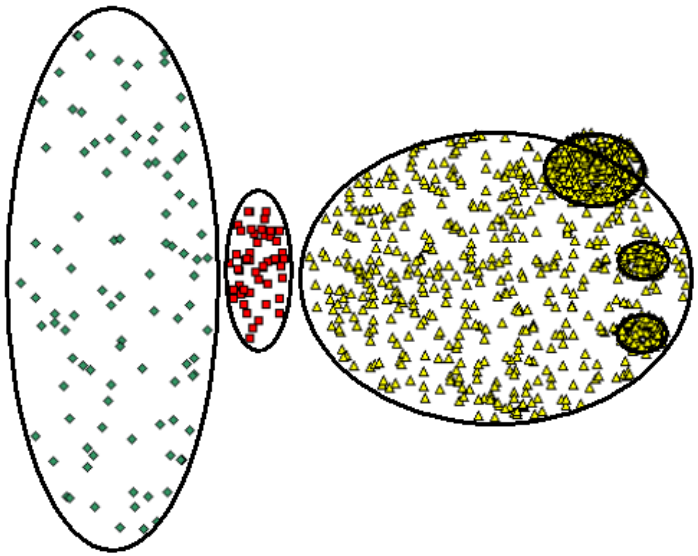


Resulting clusters

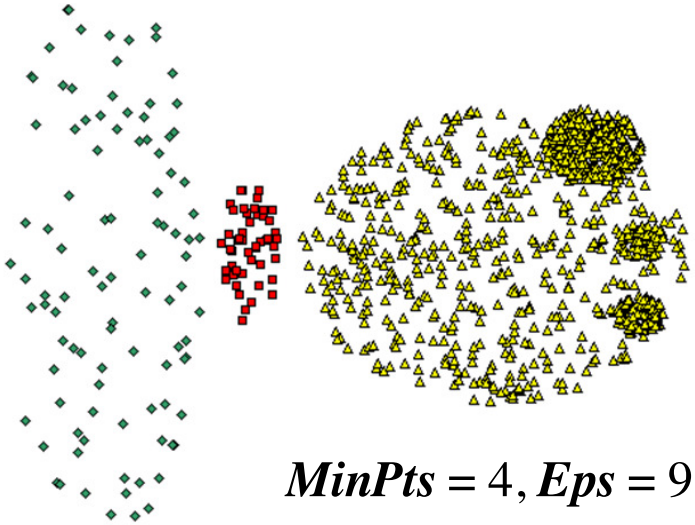
- Resistant to noise
- Can handle non-globular clusters

DBSCAN: weak points

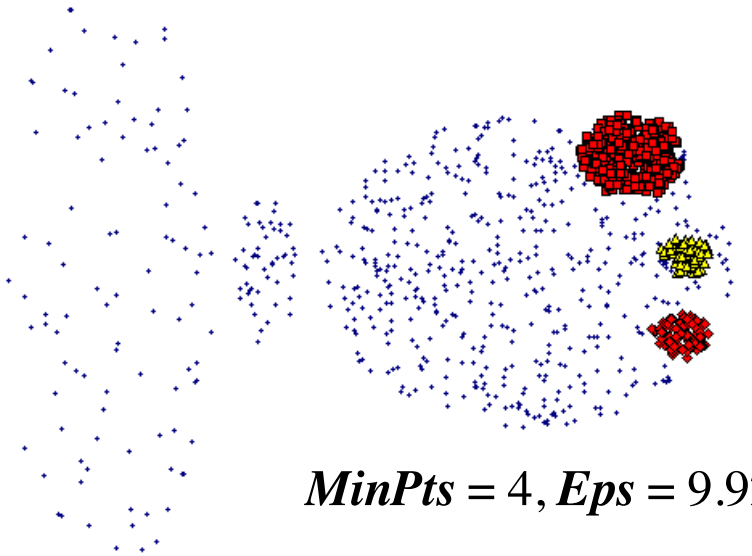
- Varying densities
- High-dimensional data



Original Points



$MinPts = 4, Eps = 9.75$



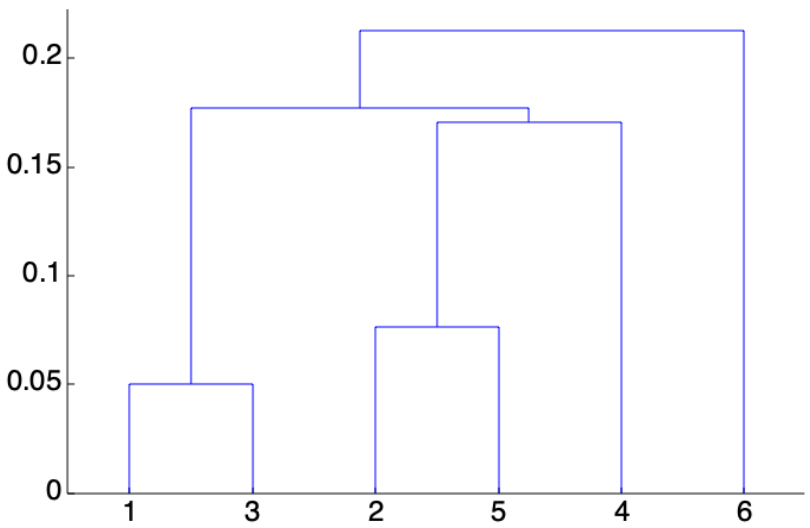
$MinPts = 4, Eps = 9.92$

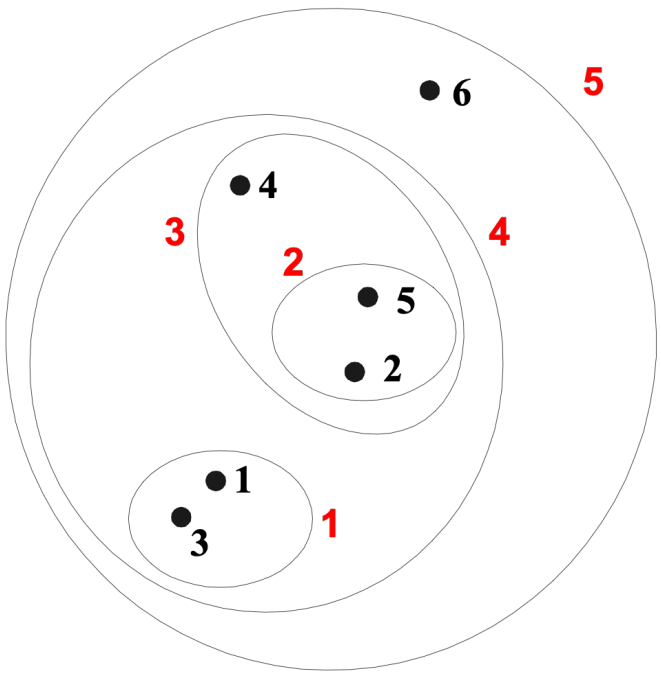
Topics

- What is Cluster Analysis
- Types of Clustering
  - Partitional: K-means
  - Density-based: DBSCAN
  - Hierarchical: Agglomerative Nesting
- Cluster Validity
  - Similarity Metric

Hierarchical Clustering

- Produces a set of nested clusters organised as a hierarchical tree
- Can be visualised as a ***dendrogram***
  - A tree like diagram that records the sequences of merges or splits





Hierarchical Clustering

- Two main types of hierarchical clustering
  - Agglomerative (bottom up):
    - Start with ***points as individual clusters***
    - At each step, merge the closest pair of clusters until a single cluster is (or ***k*** clusters are) obtained.
  - Divisive (top down):
    - Start with a single, all-inclusive cluster
    - At each step, split a cluster until each cluster contains a single point (or until there are ***k*** clusters)
- Traditional hierarchical algorithms use a similarity or a distance matrix
  - Merge or split one cluster at a time

Strengths of Hierarchical Clustering

- Do not have to assume any particular number of clusters
  - Any desired number of clusters can be obtained by 'cutting' the dendrogram at the proper level

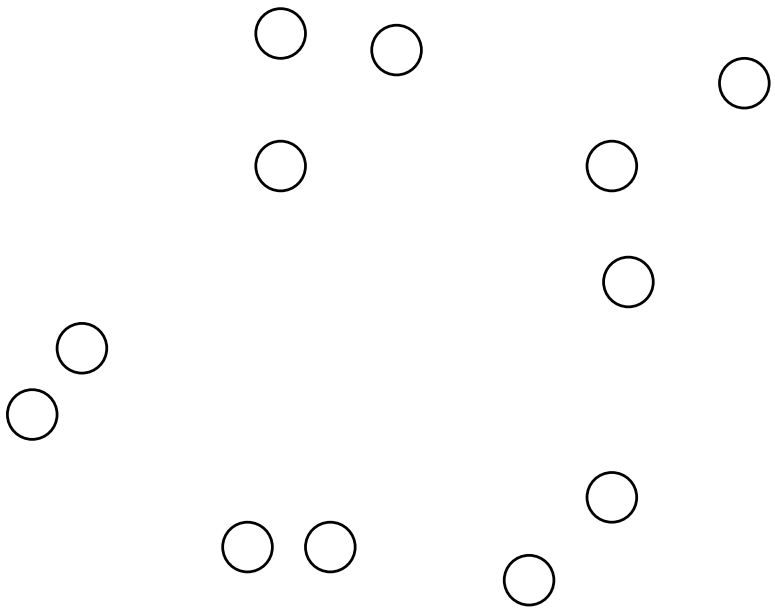
- May correspond to meaningful taxonomies
  - E.g., phylogenetic reconstruction (evolutionary trees)

Agglomerative Clustering Algorithm

- More popular hierarchical clustering technique
- Basic algorithm is straightforward:
  - Compute the proximity matrix
  - Let each data point be a cluster
  - Repeat until only a single cluster remains:
    - Merge the two *closest* clusters
    - Update the proximity matrix
- Key operation is the computation of the proximity of two clusters
  - Different approaches to defining the distance between clusters distinguish different algorithms

Starting Situation

- Start with clusters of individual points and a proximity matrix

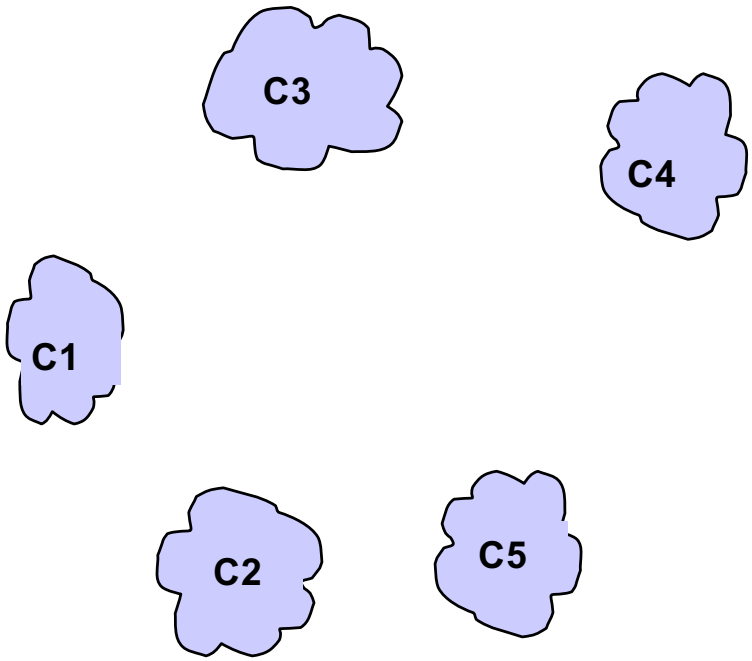


	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

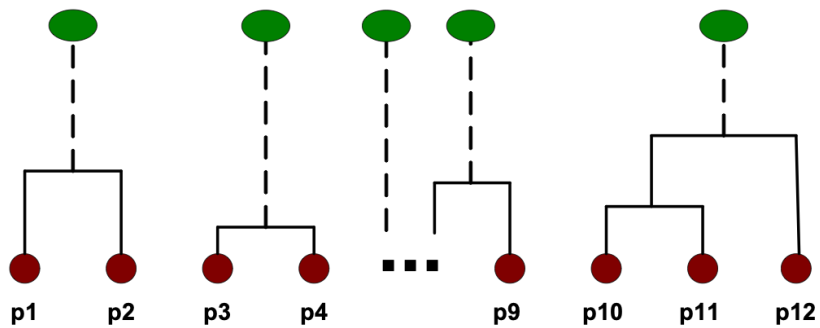
Intermediate Situation

- After some merging steps, we have some clusters



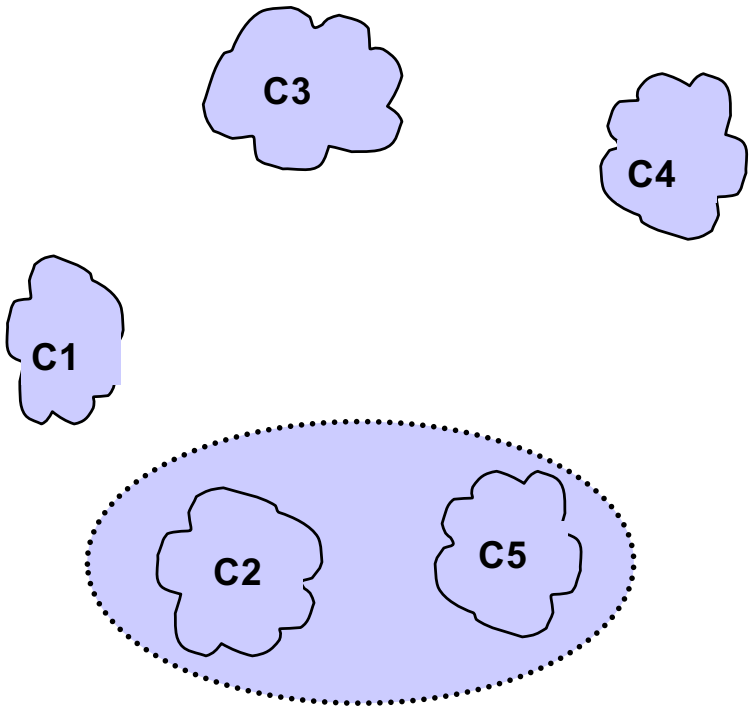
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

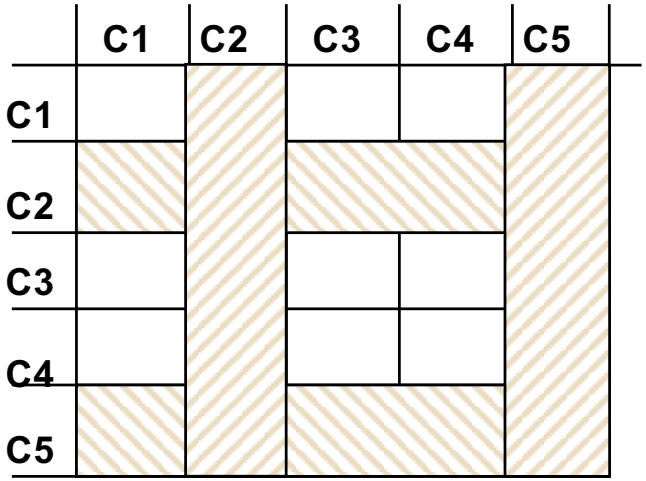
Proximity Matrix



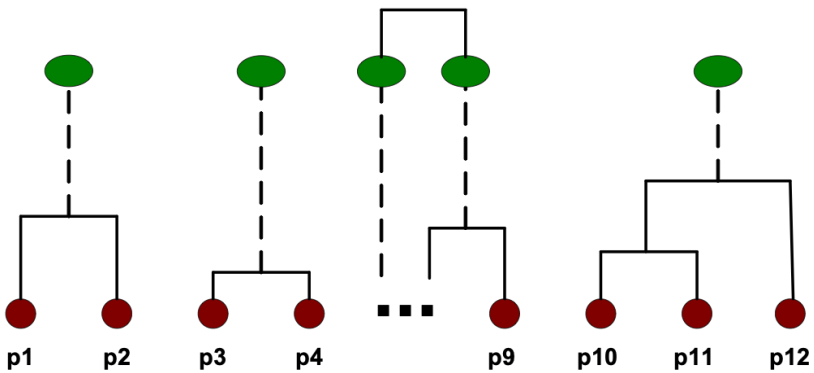
Intermediate Situation

- We want to merge the two closest clusters and update the proximity matrix.



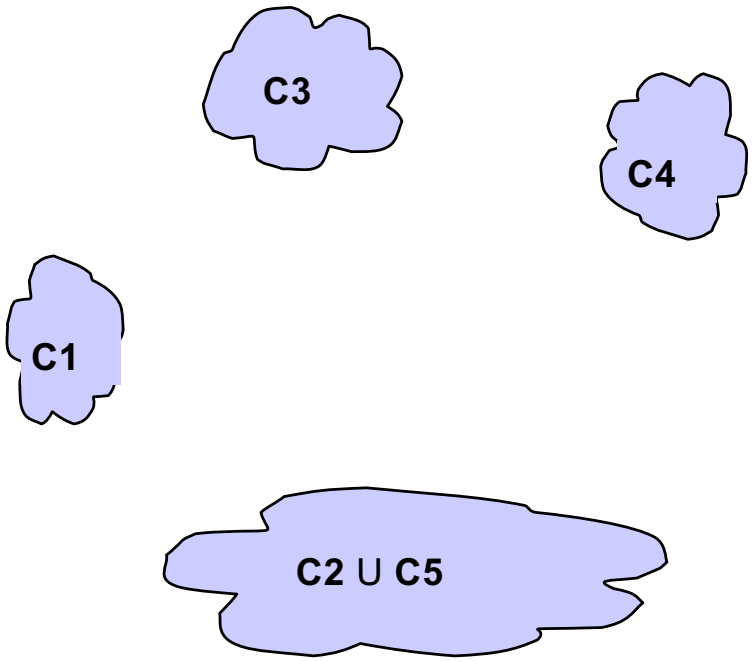


Proximity Matrix



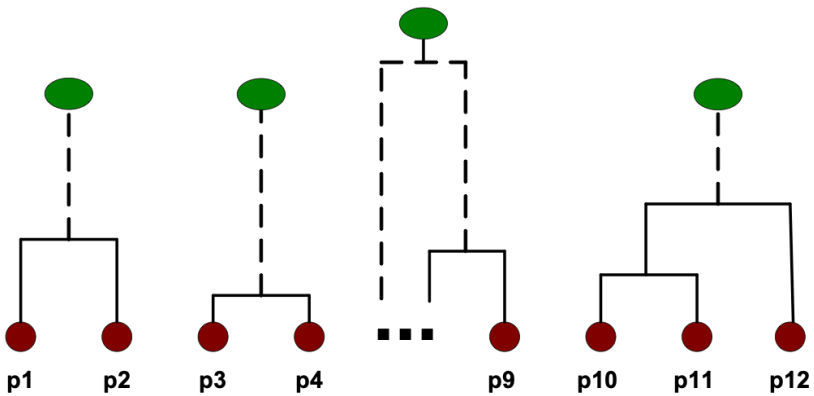
After Merging

- The question is "*How do we update the proximity matrix?*"

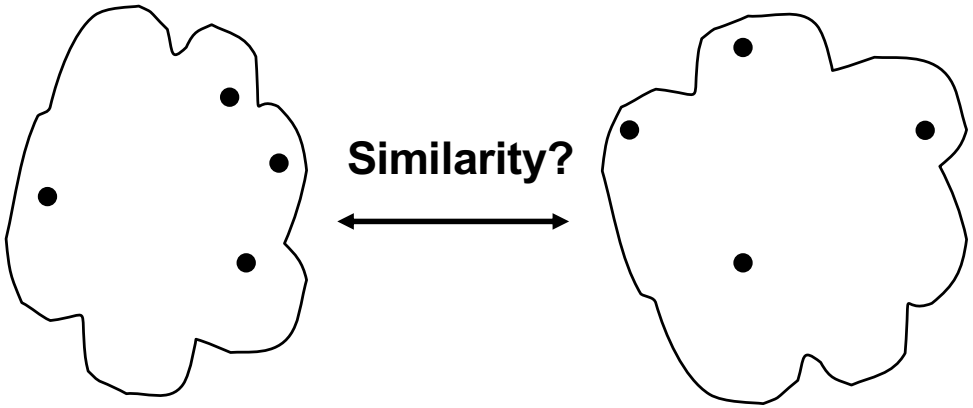


	C1	C2 U C5	C3	C4
C1		?		
C2 U C5	?	?	?	?
C3		?		
C4		?		

Proximity Matrix



How to Define Inter-Cluster Similarity

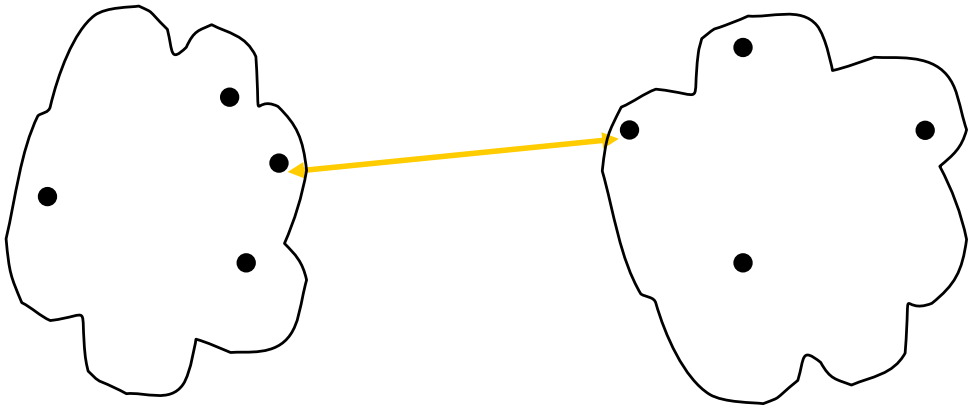


	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

- MIN
- MAX
- Group Average
- Distance Between Centroids
- …

How to Define Inter-Cluster Similarity



	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

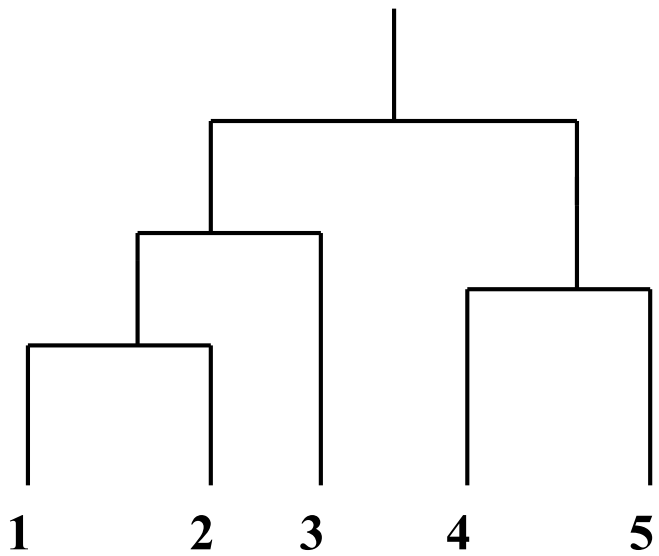
Proximity Matrix

- **MIN**
- MAX
- Group Average
- Distance Between Centroids
- …

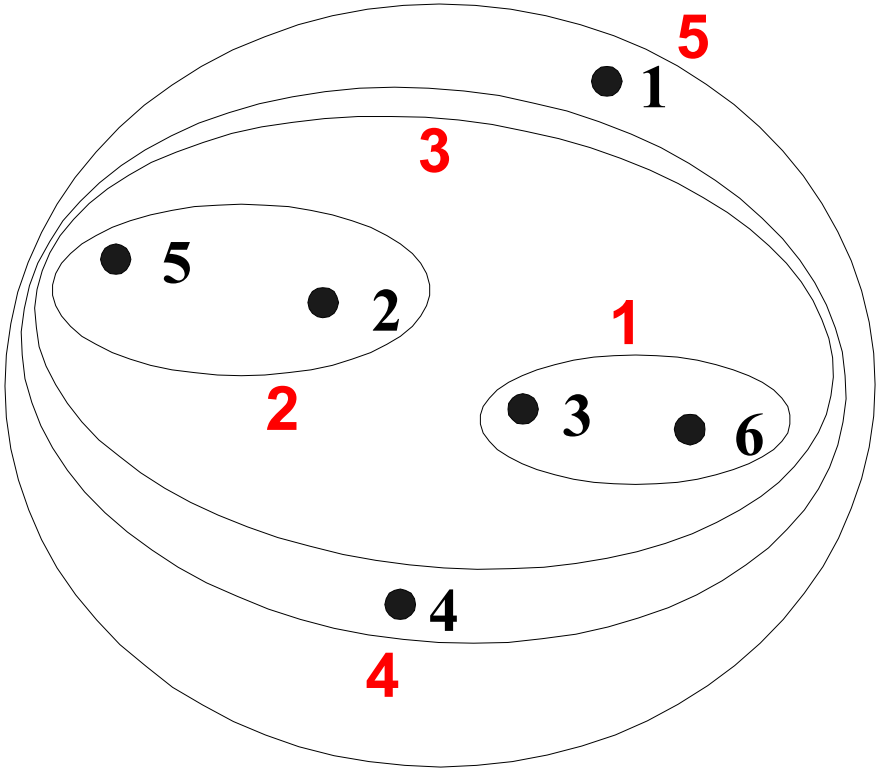
Cluster Similarity: MIN (Single Linkage)

- Similarity of two clusters is based on the two most similar (closest) points in the different clusters
  - Determined by one pair (strongest link) of points
    - i.e., by one link in the proximity graph.

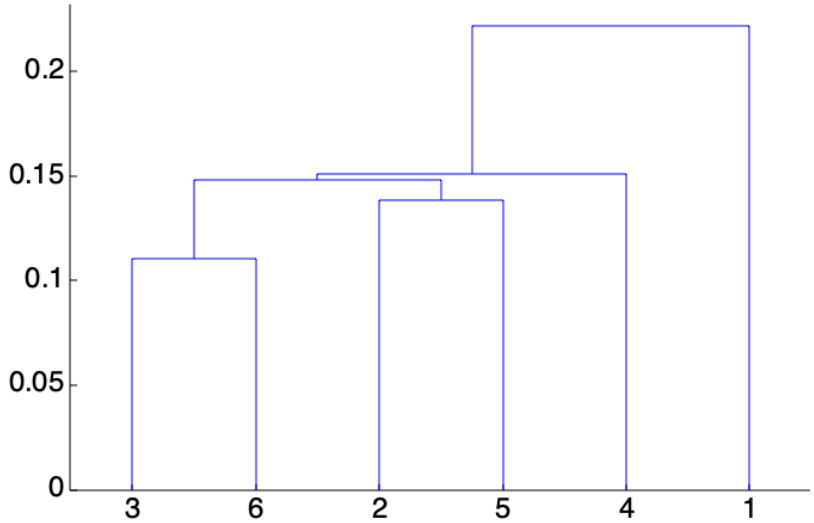
	I1	I2	I3	I4	I5
I1	1.00	0.90	0.10	0.65	0.20
I2	0.90	1.00	0.70	0.60	0.50
I3	0.10	0.70	1.00	0.40	0.30
I4	0.65	0.60	0.40	1.00	0.80
I5	0.20	0.50	0.30	0.80	1.00



Hierarchical Clustering: MIN



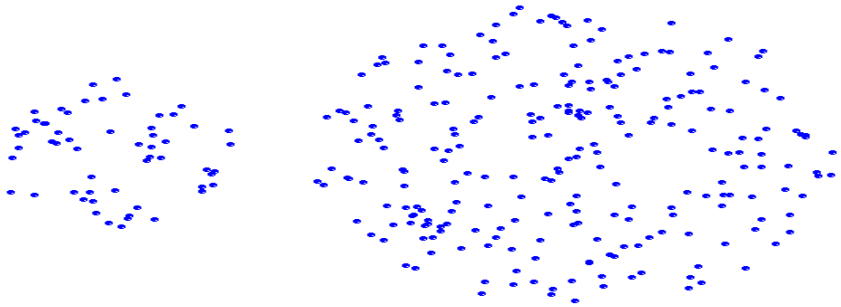
Nested Clusters



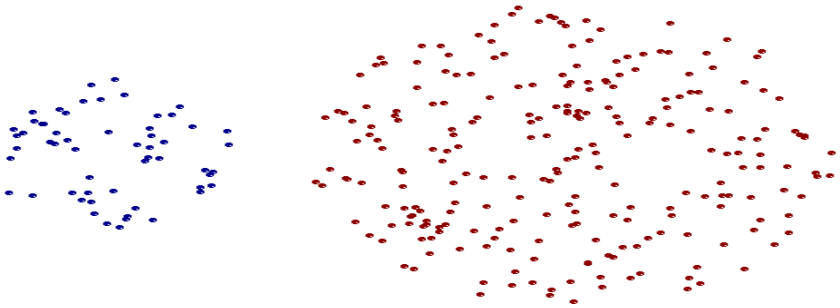
Dendrogram

Strength of MIN

- Can handle non-globular shapes



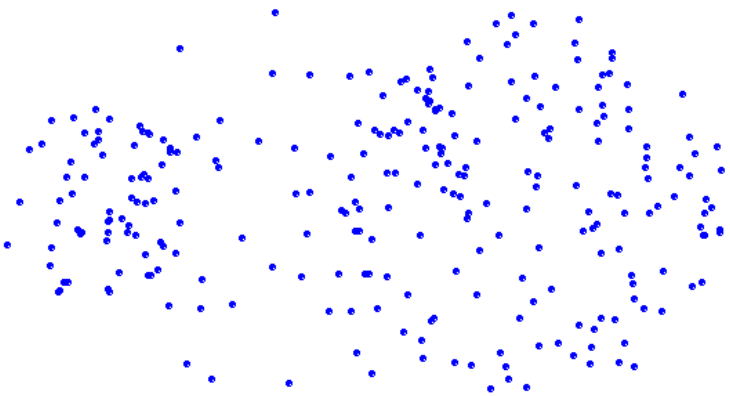
Original Points



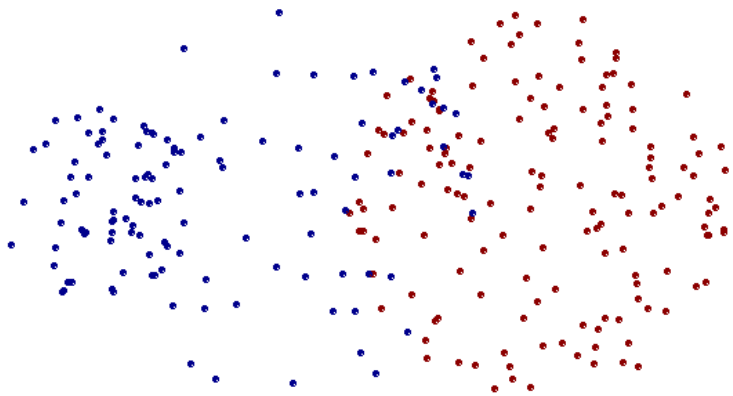
Two Clusters

Limitations of MIN

- Sensitive to noise and outliers

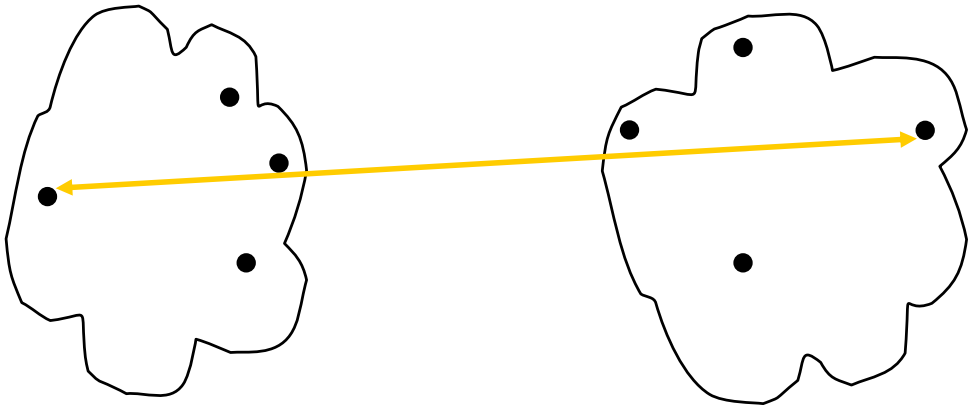


Original Points



Two Clusters

How to Define Inter-Cluster Similarity



	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

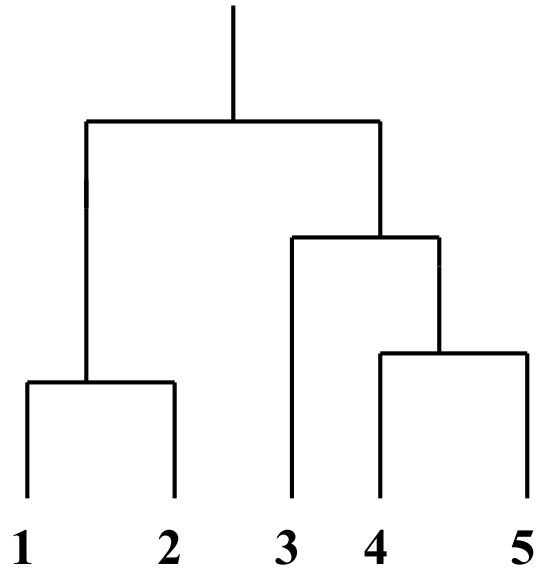
Proximity Matrix

- MIN
- **MAX**
- Group Average
- Distance Between Centroids
- …

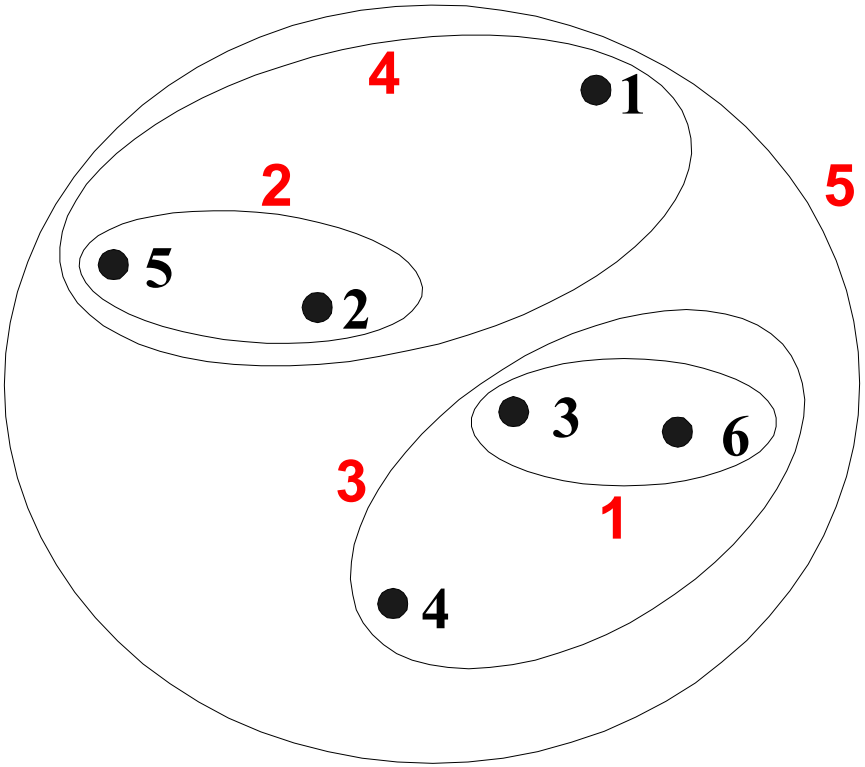
Cluster Similarity: MAX (Complete Linkage)

- Similarity of two clusters is based on the two ***least similar*** (most distant) points in the different clusters
  - Determined by all pairs of points in the two clusters
    - A cluster is formed when all points in the cluster are completely linked!

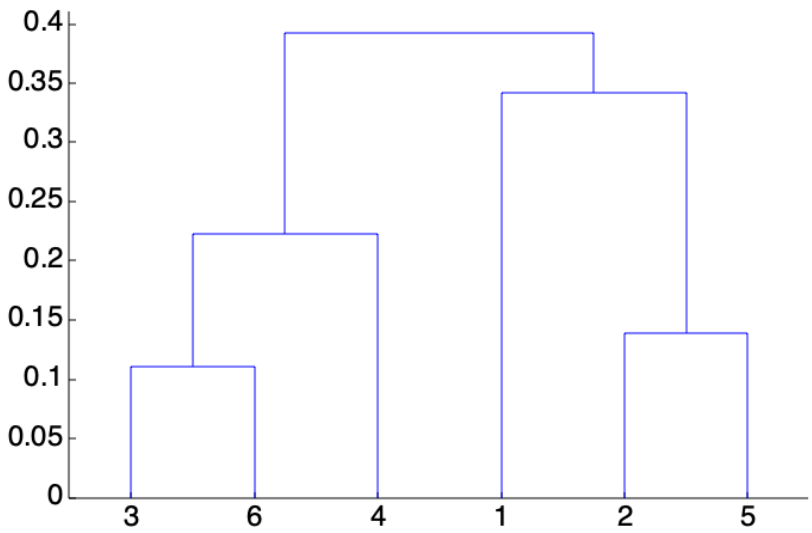
	I1	I2	I3	I4	I5
I1	1.00	0.90	0.10	0.65	0.20
I2	0.90	1.00	0.70	0.60	0.50
I3	0.10	0.70	1.00	0.40	0.30
I4	0.65	0.60	0.40	1.00	0.80
I5	0.20	0.50	0.30	0.80	1.00



Hierarchical Clustering: MAX



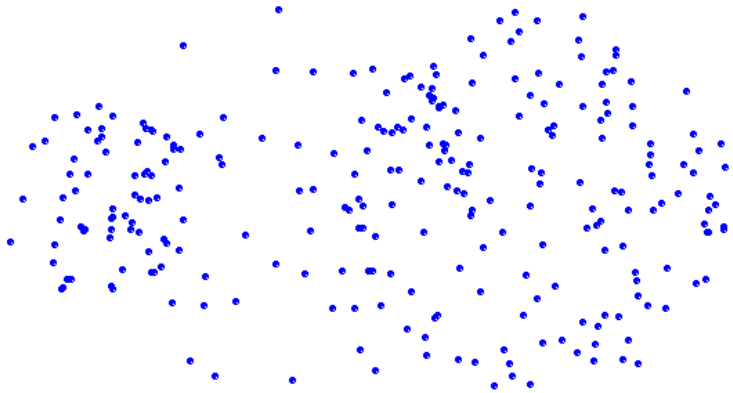
Nested Clusters



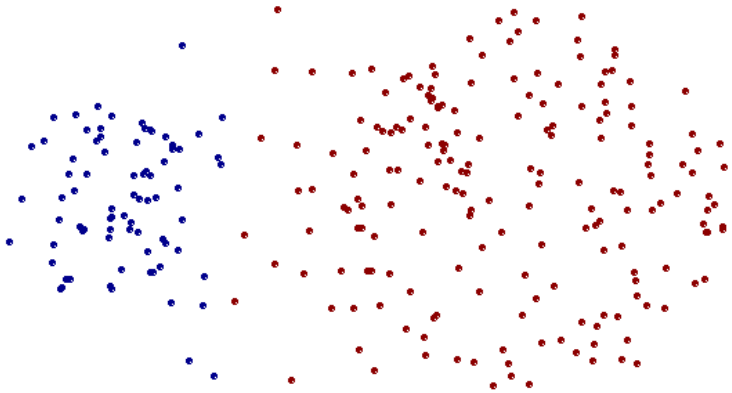
Dendrogram

Strength of MAX

- Less sensitive to noise and outliers



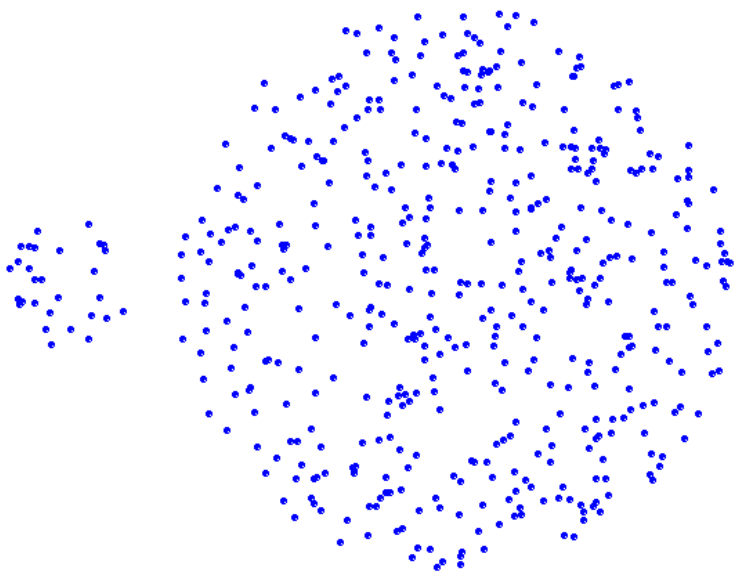
Original Points



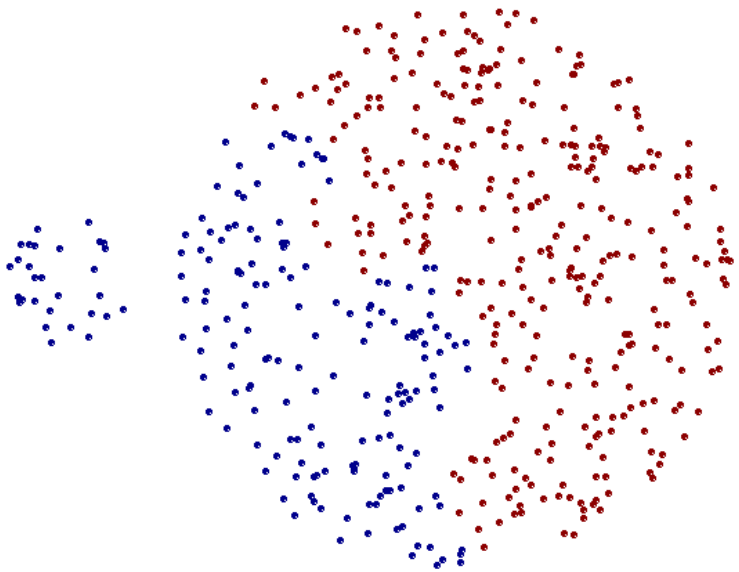
Two Clusters

Limitations of MAX

- Tends to break large clusters
- Biased towards globular clusters

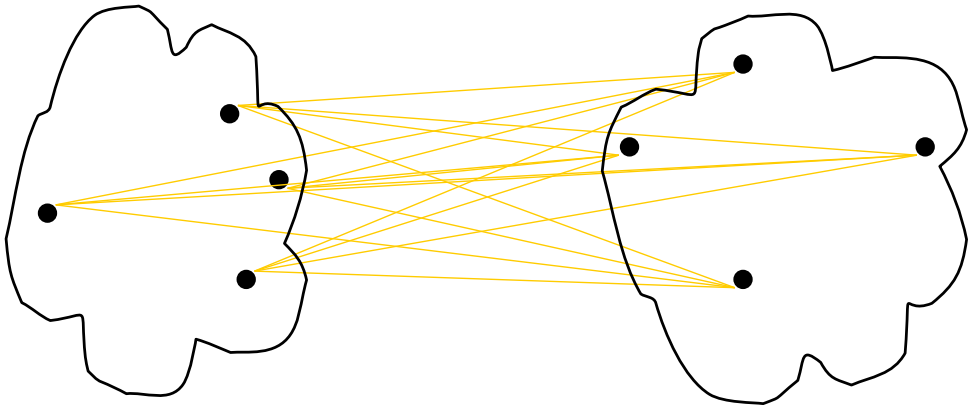


Original Points



Two Clusters

How to Define Inter-Cluster Similarity



	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

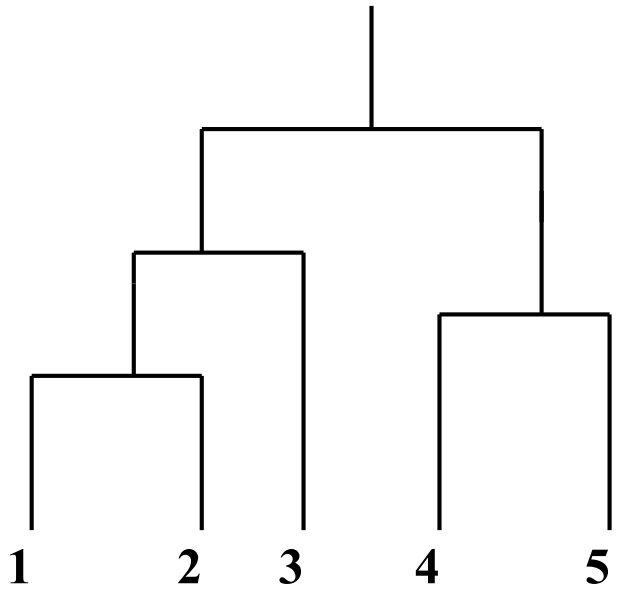
- MIN
- MAX
- **Group Average**
- Distance Between Centroids
- …

Cluster Similarity: Group Average

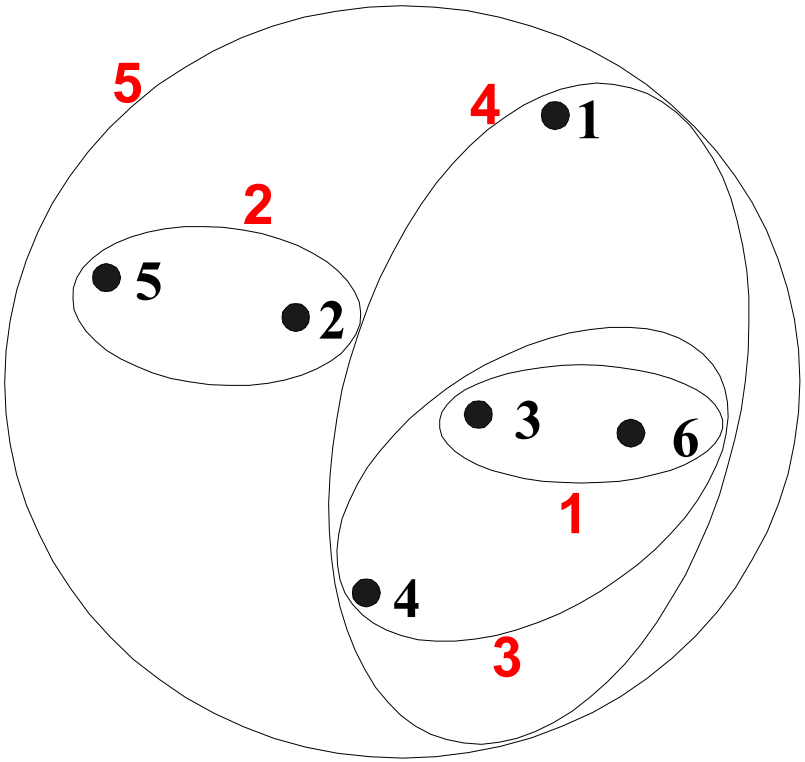
- Proximity of two clusters is the ***average of pairwise proximity*** between points in the two clusters.

$$\bar{d}(C_i, C_j) = \frac{1}{N_i N_j} \sum_{\forall x \in C_i, y \in C_j} dist(x, y)$$

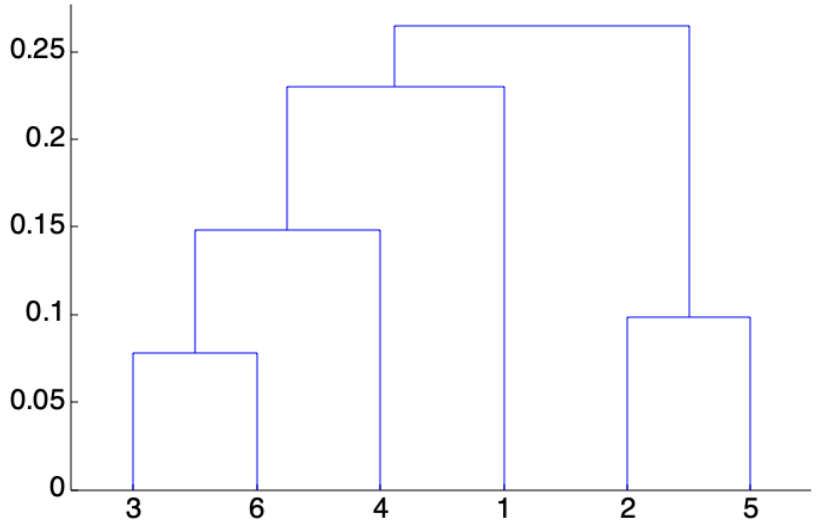
	I1	I2	I3	I4	I5
I1	1.00	0.90	0.10	0.65	0.20
I2	0.90	1.00	0.70	0.60	0.50
I3	0.10	0.70	1.00	0.40	0.30
I4	0.65	0.60	0.40	1.00	0.80
I5	0.20	0.50	0.30	0.80	1.00



Hierarchical Clustering: Group Average



Nested Clusters

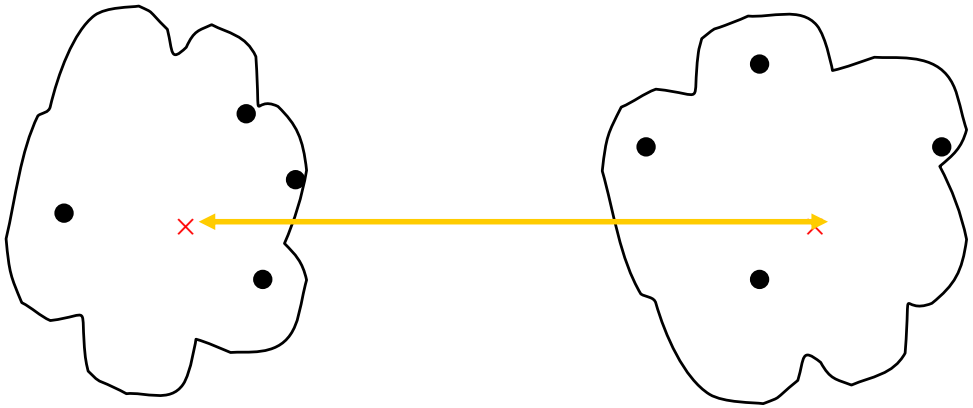


Dendrogram

Hierarchical Clustering: Group Average

- Compromise between Single and Complete Link
- Strengths
  - Less susceptible to noise and outliers
- Limitations
  - Biased towards globular clusters

How to Define Inter-Cluster Similarity



	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

- MIN
- MAX
- Group Average
- **Distance Between Centroids**
- …

Problems and Limitations

- Hierarchical clustering is not without its problems:
  - Once a decision is made to combine two clusters, it cannot be undone (no backtracking)
  - No objective function is directly minimised
  - Different schemes have problems with one or more of the following:
    - Sensitivity to noise and outliers
    - Difficulty handling different sized clusters and convex shapes
    - Breaking large clusters

Hand Calculation: MIN

- MIN (Single Linkage): five elements (a,b,c,d,e) and the similarity matrix D_1

	a	b	c	d	e
a	0	17	21	31	23
b	17	0	30	34	21
c	21	30	0	28	39
d	31	34	28	0	43
e	23	21	39	43	0

$D_1(a,b) = 17$ lowest value;

join elements *a* and *b*;

Matrix update:

$$
\begin{aligned}
D_2((a,b),c) &= \min(D_1(a,c), D_1(b,c)) &= \min(21,30) &= 21 \\
D_2((a,b),d) &= \min(D_1(a,d), D_1(b,d)) &= \min(31,34) &= 31 \\
D_2((a,b),e) &= \min(D_1(a,e), D_1(b,e)) &= \min(23,21) &= 21
\end{aligned}
$$

Hand Calculation: MIN

	(a,b)	c	d	e
(a,b)	0	**21**	**31**	**21**
c	**21**	0	*28*	*39*
d	**31**	*28*	0	*43*
e	**21**	*39*	*43*	0

$$D_2((a,b),c) = 17$$

$$D_2((a,b),e) = 17$$

join *(a, b)* with elements *c* e *e*

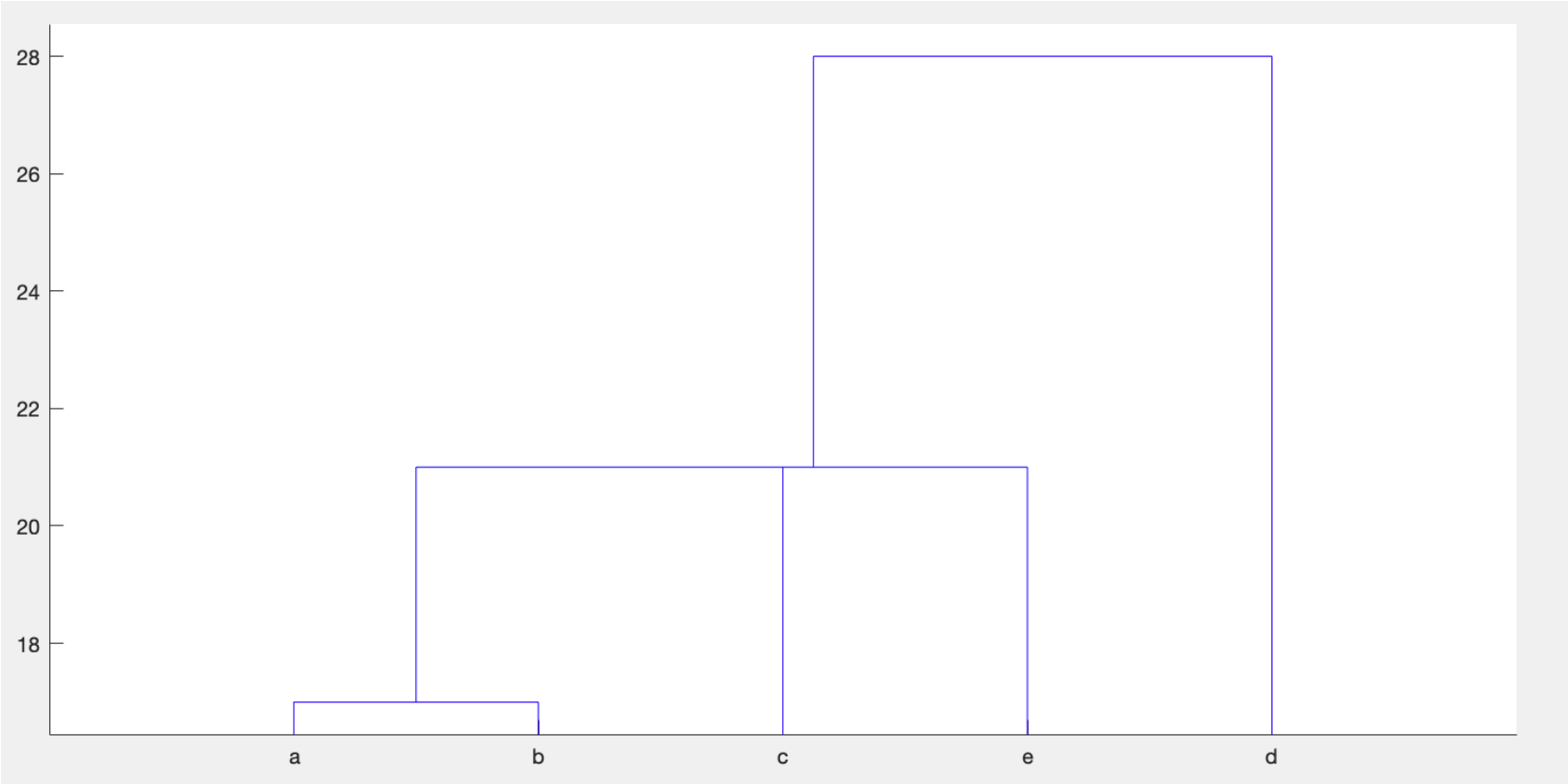
Matrix update

$$D_3(((a,b),c,e),d) = \min(D_2((a,b),d), D_2(c,d), D_2(e,d)) = \min(31, 28, 43) = 28$$

	((a,b),c,e)	d
((a,b),c,e)	0	**28**
d	**28**	0

join *((a,b), c, e)* with element *d*

Hand Calculation: MIN



Hand Calculation: MAX

MAX (Complete Linkage): five elements (a,b,c,d,e) and the similarity matrix D_1

	a	b	c	d	e
a	0	17	21	31	23
b	17	0	30	34	21
c	21	30	0	28	39
d	31	34	28	0	43
e	23	21	39	43	0

$D_1(a,b) = 17$ lowest value;

cluster *a* and *b*;

Matrix update:

$$D_2((a,b),c) = max(D_1(a,c), D_1(b,c)) = max(21,30) = 30$$

$$D_2((a,b),d) = max(D_1(a,d), D_1(b,d)) = max(31,34) = 34$$

$$D_2((a,b),e) = max(D_1(a,e), D_1(b,e)) = max(23,21) = 23$$

Hand Calculation: MAX

	(a,b)	c	d	e
(a,b)	0	**30**	**34**	**23**
c	**30**	0	*28*	*39*
d	**34**	*28*	0	*43*
e	**23**	*39*	*43*	0

$$D_2((a,b),e) = 23$$

join *(a, b)* with element **e**

Matrix update

$$D_3(((a,b),e),c) = max(D_2((a,b),c), D_2(e,c)) = max(30,39) = 39$$

$$D_3(((a,b),e),d) = max(D_2((a,b),d), D_2(e,d)) = max(34,43) = 43$$

Hand Calculation: MAX

	((a,b),e)	c	d
((a,b),e)	0	39	43
c	39	0	*28*
d	43	*28*	0

$$D_3(c,d) = 28$$

join elements *c* and *d*

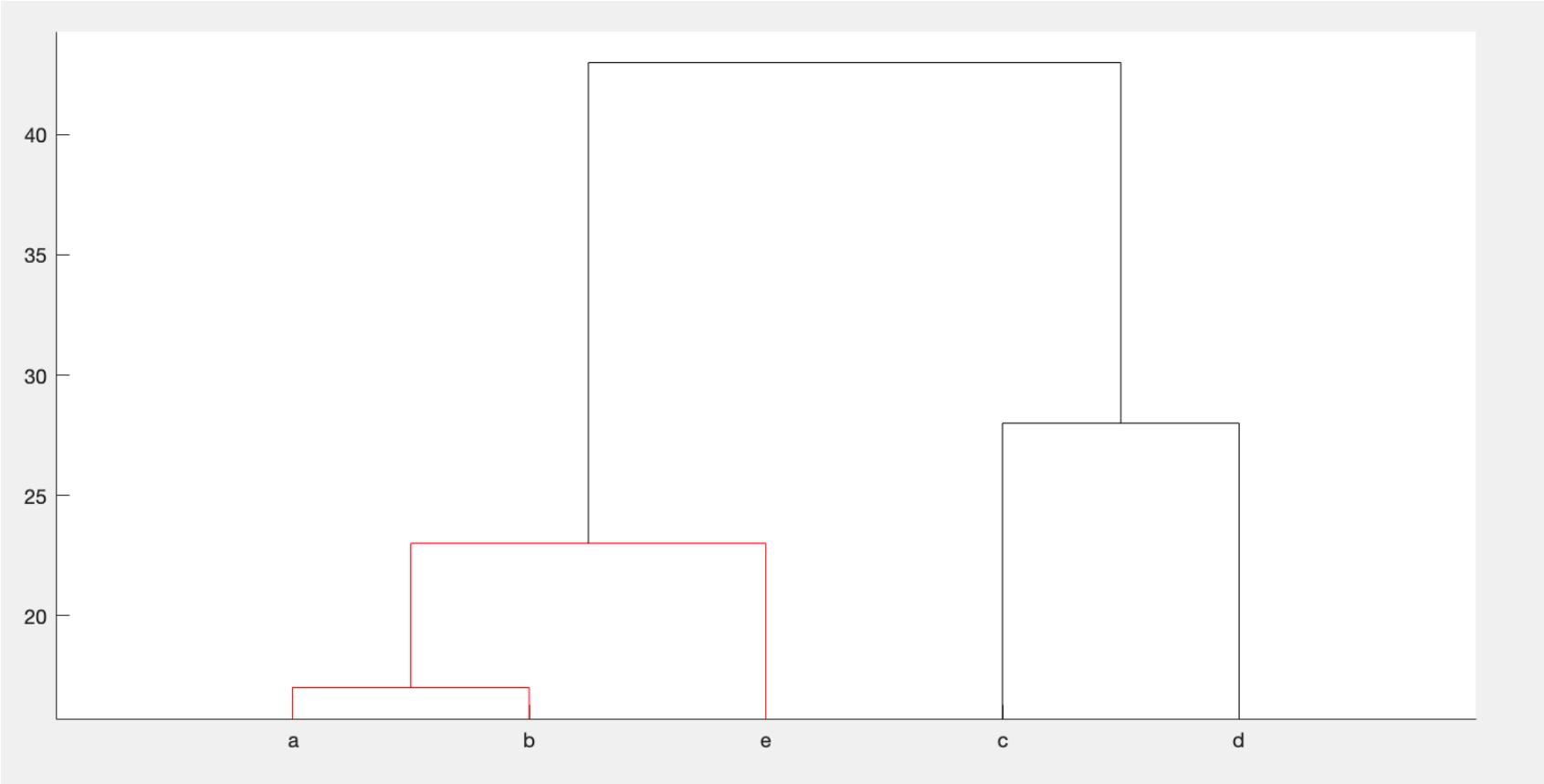
Matrix update

$$D_4((c,d),((a,b),e)) = max(D_3(c,((a,b),e)), D_3(d,((a,b),e))) = max(39,43) = 43$$

	((a,b),e)	(c,d)
((a,b),e)	0	43
(c,d)	43	0

join clusters *((a,b), e)* and *(c, d)*

Hand Calculation: MAX



Hand Calculation: Group Average

Group Average: five elements (a,b,c,d,e) and the similarity matrix D_1

	a	b	c	d	e
a	0	17	21	31	23
b	17	0	30	34	21
c	21	30	0	28	39
d	31	34	28	0	43
e	23	21	39	43	0

$D_1(a,b) = 17$ lowest value;

cluster a and b;

Matrix update:

$$D_2((a,b),c) = (D_1(a,c) \times 1 + D_1(b,c) \times 1)/(1+1) = (21+30)/2 = 25.5$$

$$D_2((a,b),d) = (D_1(a,d) + D_1(b,d))/2 = (31+34)/2 = 32.5$$

$$D_2((a,b),e) = (D_1(a,e) + D_1(b,e))/2 = (23+21)/2 = 22$$

Hand Calculation: Group Average

	(a,b)	c	d	e
(a,b)	0	**25.5**	**32.5**	**22**
c	**25.5**	0	*28*	*39*
d	**32.5**	*28*	0	*43*
e	**22**	*39*	*43*	0

$D_2((a,b),e) = 22$

join *(a, b)* with element **e**

Matrix update

$$D_3(((a,b),e),c) = (D_2((a,b),c) \times 2 + D_2(e,c) \times 1)/(2+1) = (25.5 \times 2 + 39 \times 1)/3 = 30$$

$$D_3(((a,b),e),d) = (D_2((a,b),d) \times 2 + D_2(e,d) \times 1)/(2+1) = (32.5 \times 2 + 43 \times 1)/3 = 36$$

Hand Calculation: Group Average

	((a,b),e)	c	d
((a,b),e)	0	30	36
c	30	0	*28*
d	36	*28*	0

$$D_3(c,d) = 28$$

join elements *c* and *d*

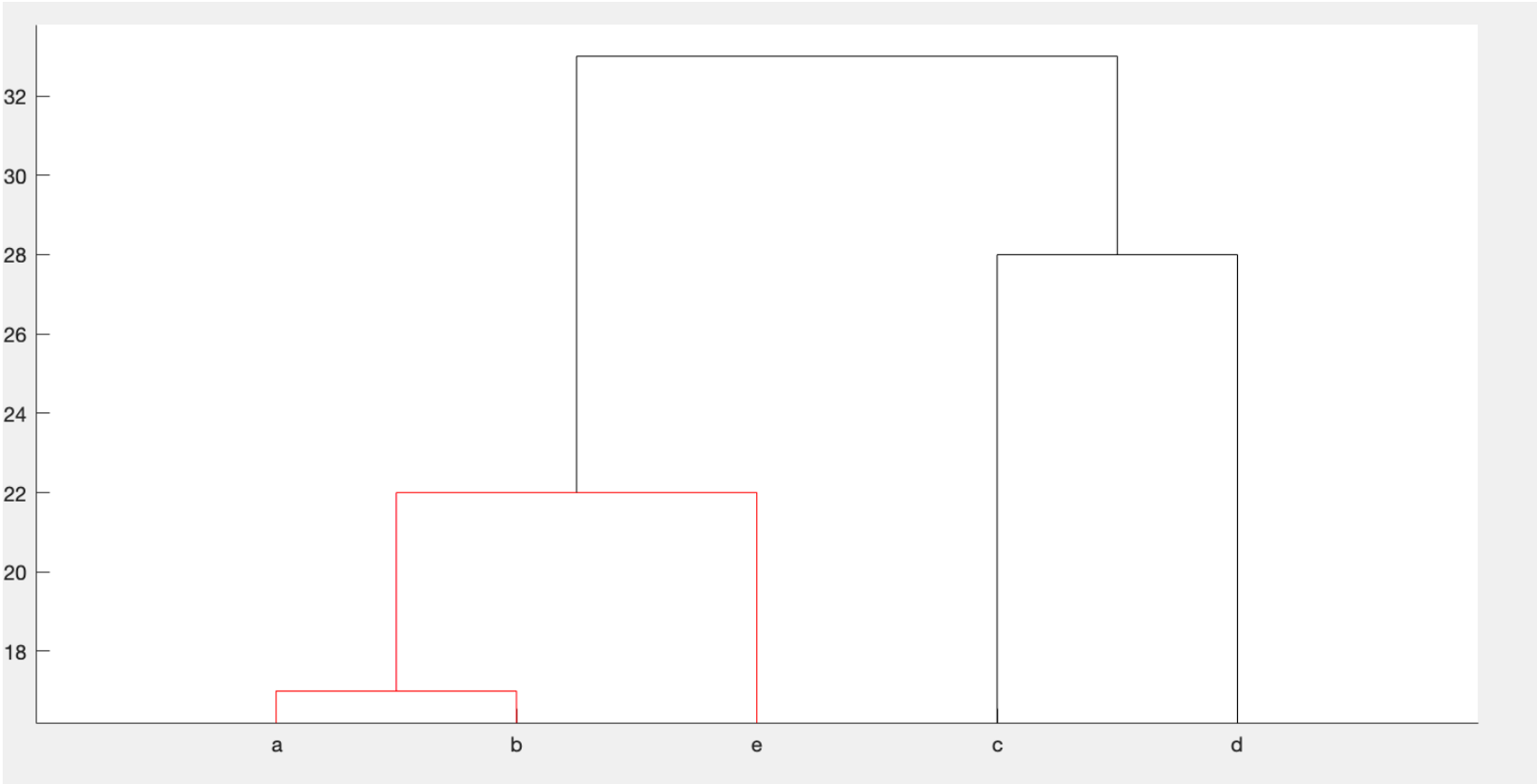
Matrix update

$$D_4((c,d),((a,b),e)) = (D_3(c,((a,b),e)) \times 1 + D_3(d,((a,b),e)) \times 1)/(1+1) = (30 \times 1 + 36 \times 1)/2 = 33$$

	((a,b),e)	(c,d)
((a,b),e)	0	33
(c,d)	33	0

join clusters *((a,b), e)* and *(c, d)*

Hand Calculation: Group Average



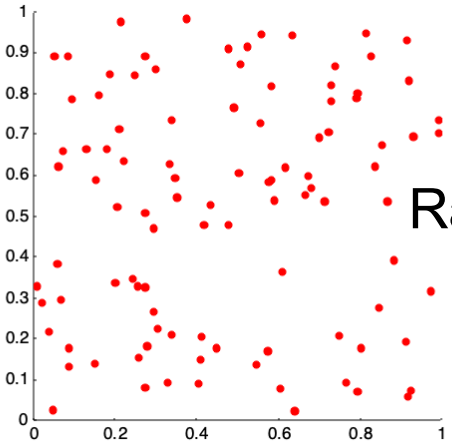
Topics

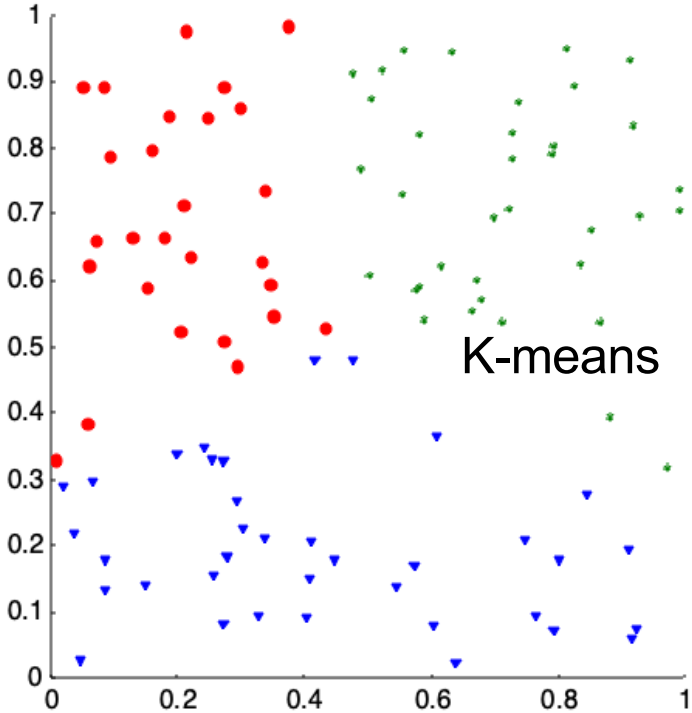
- What is Cluster Analysis
- Types of Clustering
  - Partitional: K-means
  - Density-based: DBSCAN
  - Hierarchical: Agglomerative Nesting
- Cluster Validity
  - Similarity Metric

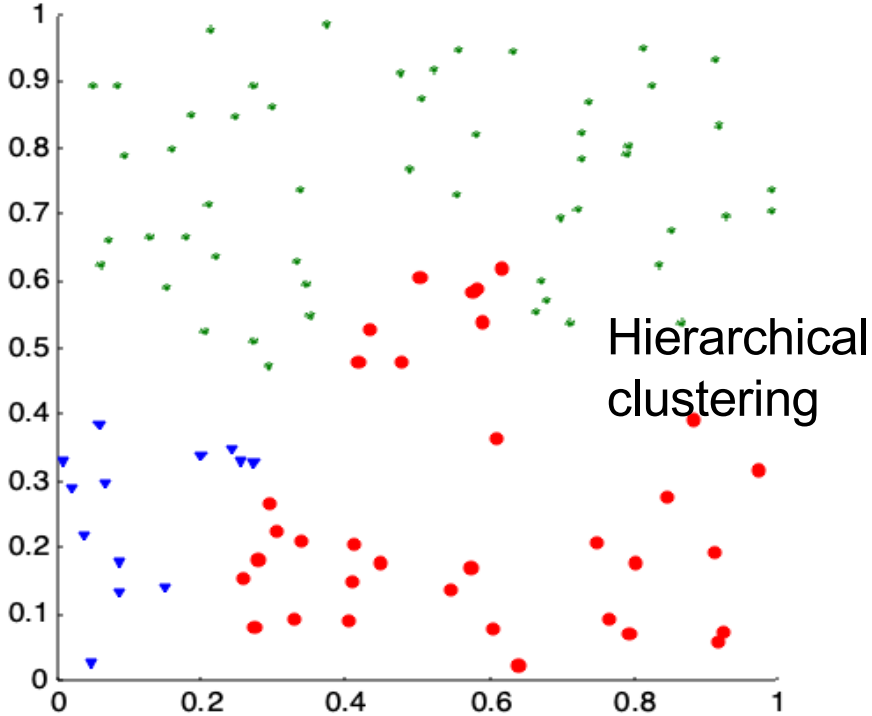
Cluster Validity

- For classification we have a variety of measures to evaluate how good our model is, e.g., accuracy, …
- For cluster analysis, the analogous question is: ***how to evaluate the "goodness" of the resulting clusters?***
- If "clusters are in the eye of the beholder", then why do we want/need to evaluate them?
  - To avoid finding patterns in noise
  - To compare clustering algorithms
  - To compare two sets of clusters
  - To compare two clusters

Clusters found in Random Data







Different Aspects of Cluster Validation

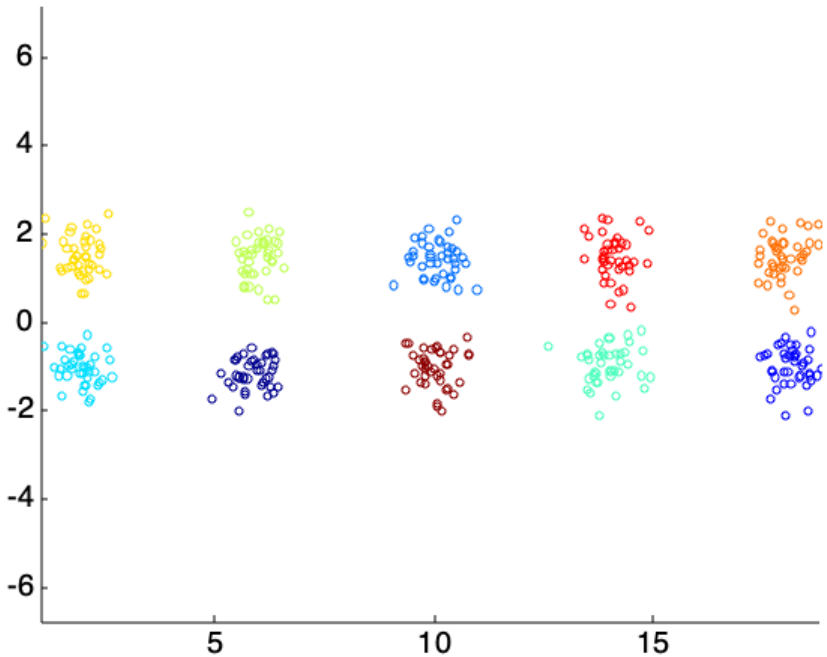
- Determining the **clustering tendency** of a data set - distinguishing whether non-random structure actually exists in the data.
- Comparing the results of a cluster analysis to externally known results, e.g., to externally given class labels.
- Evaluating how well the results of a cluster analysis fit the data without reference to external information.
- Comparing the results of two different sets of cluster analyses to determine which is better.
- Determining the 'correct' number of clusters.

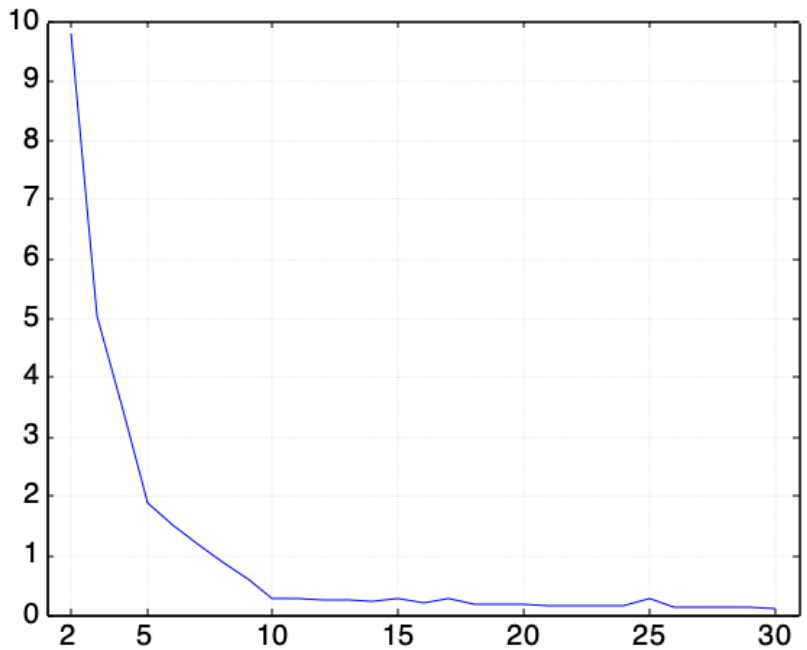
Measures of Cluster Validity

- Numerical measures are applied to judge various aspects of cluster validity:
  - **External Indices**: Used to measure the extent to which cluster labels match externally supplied class labels (e.g., entropy)
  - **Internal Indices**: Used to measure the goodness of a clustering structure without respect to external information. (e.g., SSE)
  - **Relative Indices**: Used to compare two different clusterings or clusters (often an external or internal index is used for this function, e.g., SSE or entropy)
- Sometimes these are referred to as criteria instead of indices

Internal Measures: SSE

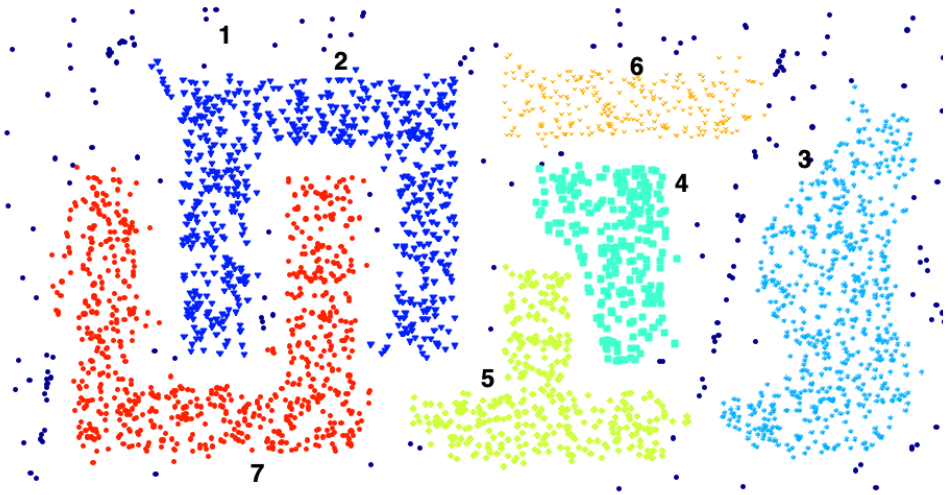
- Clusters in more complicated data aren't well separated
- Internal Index: Used to measure the goodness of a clustering structure without respect to external information
- SSE is good for comparing two clusterings or two clusters (average SSE).
- Can also be used to estimate the number of clusters

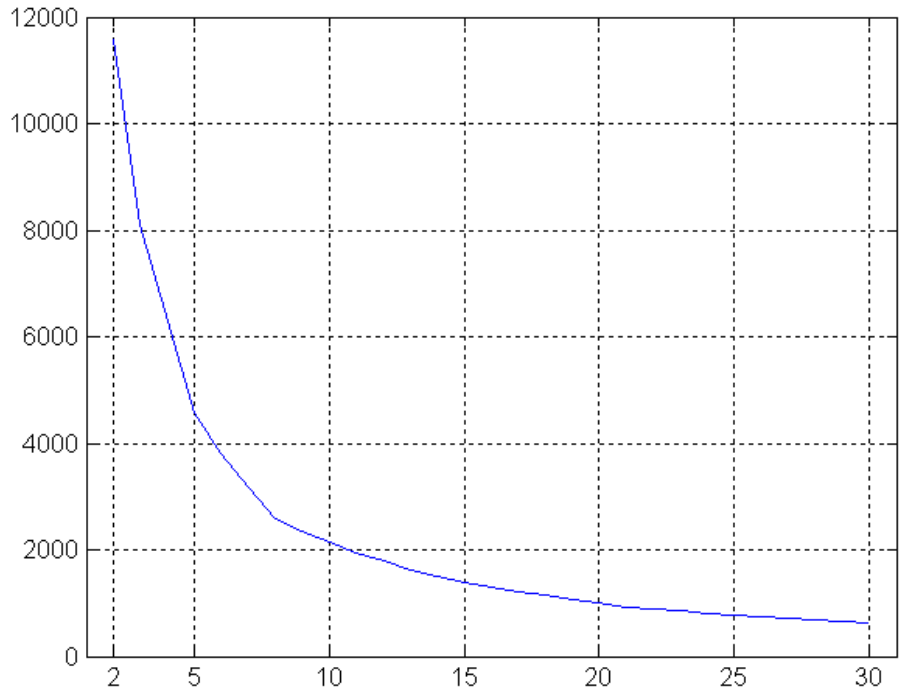




Internal Measures: SSE

- SSE curve for a more complicated data set





SSE of clusters found using K-means

Cohesion and Separation (internal)

- Cluster Cohesion: Measures how closely related objects are in a cluster, using the within-cluster sum of squares (WSS)

$$WSS = \sum_{i=1}^{K} \sum_{x \in C_i} (x - m_i)^2$$

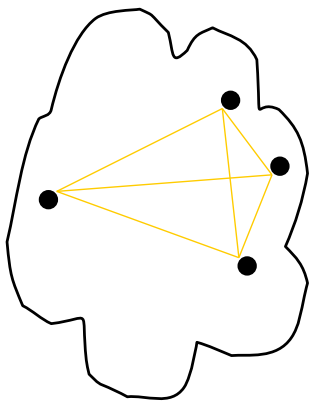
- Cluster Separation: Measures how distinct or well-separated a cluster is from other clusters, using the between cluster sum of squares (BSS)

$$BSS = \sum_{i=1}^{K} |C_i| (\overline{m} - m_i)^2$$

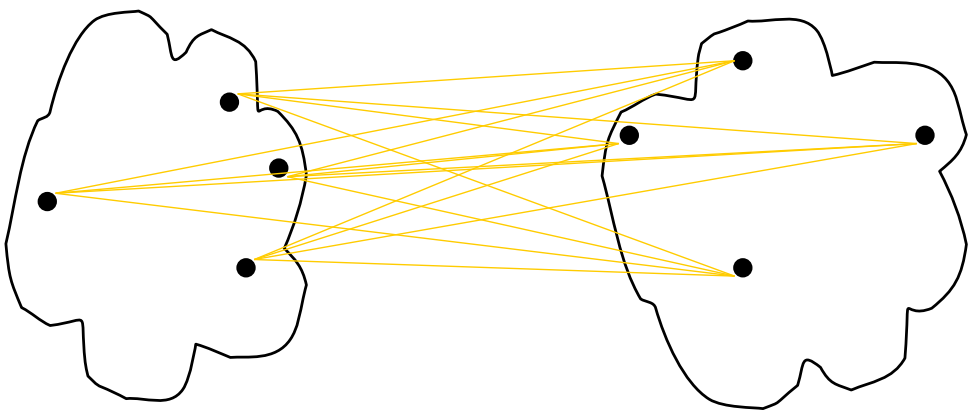
- where $\overline{m}$ is the grand mean of all centroids.

Cohesion and Separation (internal)

- A proximity graph-based approach can also be used for cohesion and separation.
  - Cluster cohesion is the sum of the weight of all links within a cluster.
  - Cluster separation is the sum of the weights between nodes in the cluster and nodes outside the cluster.



cohesion



separation

Final Comment on Cluster Validity

> The validation of clustering structures is the most difficult and frustrating part of cluster analysis. Without a strong effort in this direction, cluster analysis will remain a black art accessible only to those true believers who have experience and great courage.
>
> Jain and Dubes
> Algorithms for Clustering Data